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FINAL PROJECT REPORT

WEB DATA ANALYSIS

**ACCOUNTING AND AUDITING RECRUITMENT
MARKET ANALYSIS**

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During the process of completing this project, despite our best efforts, there may still be some limitations and shortcomings. We sincerely appreciate any comments and suggestions from lecturers and readers so that we can further improve our knowledge and develop our skills in future projects.

Best Regards./.

COMMITMENT

Our team hereby declares that this report is the result of our own research. The content, analysis results, and source code presented in this report are truthful and have never been submitted to any other coursework before.

All external sources, data, and information have been cited fully and correctly. We are solely responsible for the accuracy and truthfulness of the content in this report.

Ho Chi Minh City, March 2026

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LIST OF ABBREVIATIONS

ACCA	Association of Chartered Certified Accountants
AI	Artificial Intelligence
CPA	Certified Public Accountants
EDA	Exploratory Data Analysis
ERP	Enterprise Resource
FDI	Foreign Direct Investment
FTAs	Free Trade Agreements
ICAEW	Institute of Chartered Accountants in England and Wales
IFRS	International Financial Reporting Standards
KPIs	Key Performance Indicators
ML	Machine Learning
RPA	Robotic Process Automation
SAP	Systems, Applications, & Products
VACPA	Vietnam Association of Certified Public Accountants
VCCI	Vietnam Chamber of Commerce and Industry
WCSS	Wave Squared Distances within Clusters

ABSTRACT

This project analyzes recruitment data for accounting and auditing positions in Vietnam. The dataset was collected from the CareerViet website and includes job postings with key information such as job titles, required skills, experience levels, and salary ranges.

The data was processed and analyzed to identify recruitment trends, skill demands, and labor market characteristics in the accounting field. The findings provide valuable insights into the demand for accounting professionals and highlight the most commonly required skills and qualifications by employers.

This project aims to support students and job seekers in better understanding the accounting recruitment market and to provide practical guidance for career preparation in this field.

CHAPTER 1: INTRODUCTION

1.1. Background of the Study

Accounting and auditing are essential functions in modern organizations because they ensure financial transparency, regulatory compliance, and reliable financial reporting. These activities support decision-making for managers, investors, and regulators, making accounting and auditing professionals important contributors to business operations and financial systems.

In recent years, the labor market has undergone a significant transformation due to digitalization and the rapid development of information technologies. According to the World Economic Forum (2023), technological advancement is reshaping many occupations and changing the skill requirements expected by employers, including roles related to finance, accounting, and auditing. As a result, organizations increasingly seek candidates who possess both professional knowledge and data-related competencies.

In Vietnam, the demand for accounting and auditing professionals continues to grow as businesses expand and financial regulations become more complex. The accounting profession plays a vital role in supporting financial management, auditing activities, and compliance with accounting standards. The International Federation of Accountants highlights that professional accountants contribute significantly to the stability and transparency of financial systems in both the public and private sectors.

The growth of online recruitment platforms has generated a large volume of job market information. Job postings published on recruitment websites contain valuable data about job titles, required skills, experience levels, and salary ranges. These datasets provide an opportunity to apply data analytics techniques to better understand recruitment patterns and labor market trends.

Analyzing recruitment data from online job postings can provide useful insights into the accounting and auditing labor market. Such analysis can help students, job seekers, and educators better understand employer expectations and skill requirements, while also supporting organizations in identifying workforce trends within the industry.

1.2. Problem Statement

Despite the increasing demand for accounting and auditing professionals, organizations still face challenges in identifying candidates who possess the appropriate skills and experience. The mismatch between labor supply and labor demand creates difficulties for both employers and job seekers in the recruitment process.

Employers need a clearer understanding of the skills, experience levels, and job roles that are most demanded in the accounting and auditing labor market. Meanwhile, graduates and job seekers require better insights into the qualifications and competencies expected by employers in these fields.

Labor market information is often fragmented across different recruitment platforms. The large volume of online job postings makes it difficult to manually identify patterns or trends in recruitment activities. Without systematic data collection and analysis, valuable insights about job requirements and labor market dynamics may remain hidden.

There is a need for a data-driven analysis of recruitment data in the accounting and auditing sector. By analyzing job postings from online recruitment platforms, this study aims to identify recruitment trends, common job requirements, and skill demand patterns in the accounting and auditing labor market.

1.3. Research Objectives

The primary objective of this study is to analyze the accounting and auditing recruitment market by examining job posting data in Vietnam. The research aims to identify key trends in the demand for professionals in these fields, including commonly required skills, qualifications, and experience levels.

The study seeks to explore patterns in job roles and salary ranges within the accounting and auditing sector in order to better understand the structure of the labor market. By analyzing recruitment data, this research provides insights that may help students, job seekers, and organizations better understand current labor market dynamics and make more informed decisions regarding career development, recruitment strategies, and workforce planning.

Based on the objectives above, this study seeks to answer several key questions, including:

“What are the key trends in recruitment demand for accounting and auditing professionals in Vietnam?”

“Which skills are most frequently required by employers?”

“How are job levels and salary ranges distributed within the labor market?”

“How does experience influence salary and job opportunities?”

1.4. Scope of Study

This study focuses on analyzing recruitment activities in the accounting and auditing sector using job postings collected from the online recruitment platform CareerViet, covering all locations across Vietnam.

It should be noted that the dataset includes all available job postings obtained from the platform during the data collection process, without restricting the analysis to a specific time period. However, the study only utilizes data from the CareerViet platform and does not include recruitment information from other job portals. Hence, the findings reflect recruitment patterns based on the data available from this specific source.

1.5. Research Workflow

The research process of this study follows several main steps, starting from data collection and ending with the interpretation of analytical results. The overall workflow provides a structured approach for analyzing recruitment trends in the accounting and auditing job market.



Figure 1.1. Research Workflow

1.6. Report Structure

The research consists of 81 pages, 33 figures, and 4 tables. In addition to the introduction, conclusion, list of abbreviations, list of tables and figures, and references, the main content of the report is structured into the following chapters:

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Methodology

Chapter 4: Results and Discussion

Chapter 5: Conclusion

CHAPTER 2: LITERATURE REVIEW

2.1. Overview of the Accounting and Auditing Profession

2.1.1. Functional Significance and Economic Role of Modern Accounting and Auditing

Accounting and auditing constitute the foundational elements of financial integrity, essential for transparency and efficiency in economic systems. Accounting communicates economic information, while auditing provides independent verification and assessment. Together, they enable informed decision-making through reliable financial data.

Today, accounting and auditing are recognized as essential professional services critical to the integrity of the financial system. The evolution of this sector directly impacts the reliability of financial data, influencing resource allocation effectiveness. This profession demands a combination of technical expertise, analytical skills, and ethical commitment to safeguard stakeholder interests, ensuring dependable financial reporting for economic growth.

2.1.2. Strategic Significance of Accounting and Auditing in International Integration

Vietnam's integration into the global economy, driven by FTAs and FDI, has repositioned accounting and auditing as crucial for international trade. This evolution requires a shift where financial reporting and verification connect domestic operations with global capital markets.

The Ministry of Finance has strengthened Vietnam's adherence to IFRS through Decision 345/QD-BTC, essential for reducing informational asymmetry between local firms and global investors, ensuring financial statements are understandable and transparent. This standardization creates a demand for skilled professionals capable of complex auditing and consolidated reporting compliant with international standards.

The surge in FDI has led to a 55% increase in auditing firm clientele over five years, emphasizing the importance of verification services in attracting foreign investment. Vietnamese accountants and auditors must master both domestic tax laws and global auditing standards to meet foreign parent company requirements, maintaining competitiveness in the global market.

2.1.3. Digital Transformation and the Impact of Industry 4.0

The Fourth Industrial Revolution (Industry 4.0) signifies a pivotal transformation in the professional arena, replacing traditional manual processes with advanced automated systems. This technological shift constitutes a structural evolution that alters the essential competencies needed in contemporary recruitment.

Advancements such as RPA and AI are optimizing repetitive tasks, allowing professionals to concentrate on analytics for informed decision-making. This shift transforms accountants' and auditors' functions from data entry to strategic advisory roles, improving the interpretation of intricate financial data for practical insights.

Studies on the Vietnamese labor market indicate a notable deficiency in the "digital mindset" among entry-level applicants, despite their awareness of technological advancements. To stay competitive in an automated landscape, emerging professionals must master integrated systems like SAP/ERP and specialized software such as Misa or CaseWare for digital auditing. The shift to Industry 4.0 demands a workforce that is technically skilled in accounting standards and proficient in utilizing data-driven tools to maintain the accuracy and integrity of financial information in a real-time digital framework.

2.1.4. Current State of Human Resources and Professional Development

The Vietnamese accounting and auditing labor market faces a "Quantity versus Quality" dichotomy, as workforce growth does not align with the competencies required by modern financial services.

The rise of professionals with international certifications like ACCA and CPA Australia signifies a national effort to achieve global standards and bolster the reputation of Vietnamese auditors, with over 13,000 individuals employed in auditing as of late 2020.

Despite the influx of graduates, organizations struggle to attract high-caliber individuals skilled in complex advisory and risk management, exacerbated by seasoned auditors shifting to corporate roles, resulting in a "brain drain" that impacts senior-level expertise and market dynamics.

2.1.5. Legal Framework and Professional Ethics

The accounting profession's integrity relies on a robust legal framework and ethical mandates that promote market transparency.

Regulated by the Law on Accounting (2015) and the Law on Independent Auditing (2011), these statutes establish a framework that supports sustainable market development by defining practitioners' obligations and technical standards, ensuring alignment with national economic security and international reporting requirements.

Ethics, as the foundation of the auditing profession, safeguards public trust, with ethical training adapting to include issues like data privacy and cybersecurity; the Ministry of Finance stresses that professional skepticism and integrity supersede technical proficiency to maintain transparency in the accounting profession.

2.2. Job Market Analytics

2.2.1. Conceptual Framework and the Evolution of Labor Data

Job Market Analytics represents a sophisticated field of study that utilizes data science and statistical methodologies to examine the intricate dynamics of labor supply and demand. Traditional labor market investigations have historically depended on archival surveys or demographic census data, which often exhibit considerable temporal delays and restricted sample sizes. Conversely, modern analytical methods utilize "Big Data" from digital recruitment platforms to provide real-time insights into employer demands.

Askitas and Zimmermann (2015) argue that online data sources offer a more timely and nuanced view of economic changes than traditional statistics; in accounting and auditing, this entails extracting information from job descriptions to discern emerging professional trends.

By leveraging high-frequency data from digital platforms, researchers can detect subtle shifts in skill requirements and salary trends that conventional surveys may miss.

2.2.2. Applications in Job Demand and Structural Analysis

The primary application of Job Market Analytics involves quantifying the volume and nature of employment opportunities within a specific industrial vertical. For the Vietnamese accounting and auditing market, this analysis reveals a significant

correlation between global economic integration and the expansion of professional service requirements.

Expert analysis by Mai (2021) indicates a significant growth in the demand for accounting and auditing services in the FDI sector, approximately 55% over five years. Demand analytics facilitate the categorization of opportunities into hierarchical levels: intern, junior, senior, and management. This classification is crucial for pinpointing labor supply "bottlenecks" and enables educational institutions to adapt their curricula to meet the technical demands of various career stages. Additionally, such analytics provide insight into geographical demand, elucidating the concentration of job vacancies in major economic centers like Ho Chi Minh City and Ha Noi.

2.2.3. Salary Analytics and Economic Determinants

Salaries in accounting and auditing are influenced by various strategic factors rather than solely by seniority. Recent research indicates that FDI firms and prominent "Big 4" firms offer compensation packages approximately 20-30% above the industry average to attract skilled professionals. Additionally, regional economic conditions significantly impact salaries, with major cities offering higher wages to accommodate living expenses and competition for talent. Modern Job Market Analytic evaluates the economic value of specific professional skills, where obtaining international certifications or expertise in systems like SAP and Oracle leads to considerable salary advancements compared to those with only local software proficiencies. This shift highlights a market evolution towards "value-based" remuneration, where technical specialization is the main driver of income advancement.

2.2.4. Skill Demand Analysis and the Digital Competency Gap

The analysis of skill demand represents an essential element of Job Market Analytics in the context of Industry 4.0. This analytical approach entails the juxtaposition of competencies enumerated in job postings with prevailing workforce profiles to pinpoint "Skill Gaps."

Current market intelligence reveals a significant reduction in the demand for manual bookkeeping proficiencies, accompanied by a notable increase in the requirement for data analytical skills. Research disseminated in Economy and Forecast

(2024) underscores the necessity for Vietnamese accounting students to cultivate a "Digital Mindset" to proficiently utilize specialized software in a digital economic landscape. This transformation compels a transition from conventional technical accounting expertise to a hybrid paradigm that encompasses data visualization, financial modeling, and proactive risk assessment.

2.2.5. Methodological Justification for the Research Analytical Model

The research framework utilizes a Binary Skill Matrix methodology based on an analysis of 959 job advertisements. This model offers various analytical benefits.

Analyzing extensive recruitment data reduces subjective biases common in qualitative research. The investigation's examination of nearly one thousand data points ensures a representative and statistically significant depiction of market demand.

The framework allows quantitative evaluation of software applications, such as Misa, Excel, and SAP, alongside key competencies like Tax Settlement and Financial Reporting. This method enables a clear distinction between "essential" and "desirable" skills in the Vietnamese labor market.

In contrast to prior studies that have largely focused on broad theoretical constructs of general skills, this model rigorously investigates the empirical interactions between clusters of technical competencies and the necessary qualifications linked to specific job classifications. This permits a more nuanced comprehension of the ways in which skill requirements transition as an individual advances from an Intern or Junior role to Senior and Managerial positions.

2.3. Related Studies

2.3.1. Synthesis of Previous Research on Accounting and Auditing Competencies

Prior research in accounting and auditing has primarily aimed to define essential skills for novice professionals. Initial investigations highlighted the "Expectation Gap" between academic instruction and practical workplace requirements.

Mai (2021) asserts that while Vietnam's accounting and auditing sector has expanded notably, it still encounters quality challenges in human resources amid international integration. Conventional studies have typically divided professional competencies into technical knowledge and "soft skills" like communication, ethics, and skepticism. Nonetheless, the rise of the digital economy has prompted a reassessment of

these classifications. Contemporary studies contend that the traditional auditor is evolving into a "hybrid professional," tasked with ensuring financial accuracy while managing intricate information systems.

2.3.2. Technological Impacts and Student Perceptions in the Vietnamese Context

Contemporary domestic research initiatives have commenced investigations into the specific ramifications of Industry 4.0 on the cognitive frameworks of the forthcoming labor force in Vietnam. A notable scholarly article published in *Economy and Forecast* (2024) examined the perceptions held by students specializing in accounting and auditing concerning the socio-technological transformations occurring in society.

This study highlights that students acknowledge digital transformation's significance but lack a profound understanding of technologies like AI and Big Data that will alter their tasks, especially in automated audit sampling and risk identification. The research findings advocate for enhancing educational curricula to foster a "Digital Mindset" rather than merely teaching theoretical concepts. This recommendation is consistent with Herath's (2024) global perspectives, which indicate that automating traditional bookkeeping and basic audit verification necessitates a transition to roles characterized by high-value analytical and forensic contributions.

2.3.3. Methodological Trends in Job Market Research

Traditionally, qualitative interviews and limited surveys have been utilized to analyze labor market trends. Reports from the Vietnam Association of Certified Public Accountants (VACPA) have provided insights into auditing firm growth but lack detailed data on technical skill requirements at various audit seniority levels.

Globally, there is a notable transition towards "Data-Driven" approaches, where researchers analyze extensive online job postings to ascertain real-time skill demands. Such evaluations reveal an increasingly dynamic auditing labor market, with the rapid obsolescence of technical skills like digital auditing standards. This methodological evolution underscores the necessity for objective and thorough data analyses to accurately reflect the contemporary landscape and shifting requirements of the accounting and auditing fields.

2.3.4. Identification of the Research Gap

This study seeks to rectify significant gaps in current academic discourse. A major issue is the dependence on subjective survey data in local studies, which may not correspond to official auditor and accountant job specifications. There is a conspicuous lack of empirical investigations employing "Big Data" or web-scraping methodologies to thoroughly examine the Vietnamese recruitment context. Additionally, while numerous studies reference "Technology," few establish a detailed statistical relationship between specific software skills in auditing and accounting and their pertinence to job levels or remuneration.

This research intends to bridge these deficiencies by employing a Binary Skill Matrix and a quantitative analysis of real job advertisements. This approach offers a precise, evidence-based representation of the existing Vietnamese accounting and auditing landscape, yielding empirical data on the connections between particular technical skill sets and professional rank.

CHAPTER 3: METHODOLOGY

3.1. Research Framework

Based on the theoretical foundation and urgency discussed in the previous chapter, the following diagram fully describes the specific implementation process of the research.

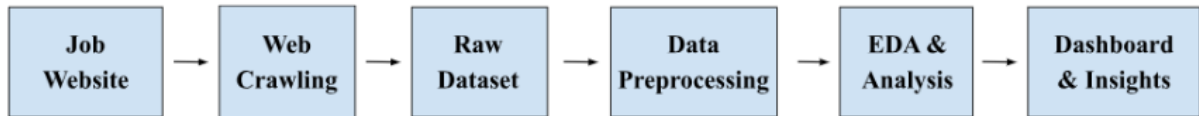


Figure 3.1. Framework Diagram

3.2. Data Collection

3.2.1. Data Source

The data source used in this study is the CareerViet recruitment platform. CareerViet is one of the major online job portals in Vietnam that provides a large number of job postings across various industries.

The dataset includes accounting and auditing job postings collected from all available locations and time periods on the platform. It contains key information such as job titles, required skills, experience levels, and salary ranges. The collected data is used to analyze recruitment trends and key characteristics of the accounting and auditing labor market.

3.2.2. Web Crawling Method

This study adopts a web crawling approach to collect job posting data from the CareerViet platform. The crawling process is implemented using Python, supported by Selenium and BeautifulSoup libraries, which enable efficient data extraction from online job portals.

Selenium is employed to automate browser interactions and handle dynamic web content. As job listings on CareerViet are partially rendered using JavaScript, Selenium allows the system to simulate user actions such as navigating pages and accessing detailed job postings, ensuring that all relevant content is fully loaded before extraction.

Once the web pages are loaded, BeautifulSoup is used to parse the HTML structure and extract key information, including job title, salary, experience requirements, and job descriptions. The integration of Selenium and BeautifulSoup

provides a robust solution for handling both dynamic and static content, ensuring that the collected data is accurate, complete, and suitable for further analysis.

3.2.3. Crawling Process

The framework of job posting data crawling of this study is presented below.



Figure 3.2. Data Crawling Process

3.2.4. Collected Variables

After the crawling process, various variables were collected from the job postings. These variables include basic job information, salary details, experience requirements, and job descriptions. Table 3.1 summarizes the variables included in the dataset and their corresponding descriptions.

Table 3.1. Collected Variables

Variable	Data Type	Description
job_id	object	Unique identifier of each job posting
job_title	object	Title of the job position
experience_min	float64	Minimum required years of experience
experience_max	float64	Maximum required years of experience
salary_min	float64	Minimum salary offered for the position
salary_max	float64	Maximum salary offered for the position
salary_string	object	Salary information displayed in the job posting
location	object	City or province where the job is based
address	object	Detailed workplace address

job_active_date	object	Date when the job posting becomes active
link	object	URL link to the detailed job posting page
job_description	object	Detailed description of the job responsibilities
requirements	object	Qualifications and requirements for the job

These variables form the foundation of the dataset used for further data processing and analysis in the subsequent chapters.

3.3. Data Preprocessing

After the data collection stage, the raw dataset requires several preprocessing steps before it can be used for analysis. Data preparation plays an important role in improving the quality, consistency, and reliability of the dataset. Since job postings collected from online recruitment platforms often contain inconsistent formats, missing values, and unstructured text information, several processing techniques are applied to standardize and transform the raw data.

3.3.1. Data Cleaning

Data cleaning is the first step in the data preparation process. The raw dataset collected from the recruitment website may contain duplicate records, incomplete information, or inconsistent data formats. Consequently, several cleaning techniques are applied to improve the overall data quality.

Duplicate job postings are identified and removed to avoid repeated observations in the dataset. Duplicate records may occur when the same job posting appears multiple times during the crawling process.

Missing values are addressed. Initially, several imputation methods, such as replacing missing values with the median for numerical variables and the mode for categorical variables, were considered to preserve as much data as possible. However, experimental results showed that these approaches led to diluted patterns and reduced model performance. So, to ensure data quality and maintain the reliability of the analysis, all records containing missing values were ultimately removed from the dataset.

These data cleaning steps help ensure that the dataset is accurate and suitable for further processing.

3.3.2. Salary Standardization

Salary information in job postings is often presented in different formats, including salary ranges, single numeric values, or textual descriptions such as “negotiable”. To make the salary information suitable for analysis, several processing steps are applied to standardize and transform the salary data.

Salary values are extracted from the textual format and converted into numerical variables. For job postings that provide salary ranges, two variables are created: minimum salary and maximum salary. Based on these values, an average salary is calculated to represent the estimated salary level of each job posting.

After normalization, an equally important step is to check the validity of the minimum and maximum salary values. Rows with minimum values greater than maximum values will be removed to ensure the data's logical consistency.

These processing steps transform the raw salary information into structured variables that can be used for further statistical analysis and visualization.

3.3.3. Experience Standardization

Experience requirements in job postings are often presented in textual formats such as “1–2 years of experience” or “at least 3 years”. To enable quantitative analysis, the required years of experience are extracted and converted into structured numerical variables.

The experience information is parsed from the job posting text. When a range of experience is provided, such as “1–3 years”, two variables are created to represent the minimum experience and maximum experience requirements. Then, an average experience value is calculated based on those values.

If experience information is not clearly specified, the value is recorded as missing or unknown. The processed experience variables are then used in subsequent analysis of recruitment demand across different experience requirements.

Similarly, the minimum and maximum values of the experience must also be checked, and any invalid lines should be removed.

3.3.4. Job Level Classification

Based on the extracted experience information, job postings are further categorized into several experience levels to facilitate analysis. In this study, the job postings are classified into six groups, as presented in Table 3.2.

Table 3.2. Job Level Mapping

Level	Experience Range (years)	Description
Unknown	Not specified	Experience information is not provided
Intern	0	Internship or entry-level positions for students
Fresher	$0 < \text{experience} \leq 1$	New employees with little or no working experience
Junior	$1 < \text{experience} \leq 2$	Employees with some initial professional experience
Senior	$2 < \text{experience} \leq 4$	Experienced employees capable of handling more complex tasks
Manager	> 4	Managerial or leadership positions

This classification simplifies the analytical process and enables comparisons of recruitment demand across different experience levels in the accounting job market.

3.3.5. Time Standardization

Job posting dates are transformed into structured time-related features to support temporal analysis. Specifically, the original date information is converted into a standardized datetime format, from which variables such as posting year and posting month are derived. These features enable the analysis of recruitment trends over time and support time-based visualizations in later stages of the study.

3.3.6. Location Standardization

Location data is processed to ensure consistency and support reliable geographical analysis. The process begins by identifying and displaying all unique location values in the dataset. This step provides an overview of how location

information is represented and helps detect inconsistencies such as variations in naming or formatting.

In cases where job postings include multiple locations, the location field is split into separate city-level entries. This allows each location to be treated individually, ensuring more precise analysis of job distribution across different areas.

City names are standardized to achieve uniformity throughout the dataset. Different variations referring to the same city are consolidated into a consistent format, which improves data quality and prevents duplication during aggregation.

To enable broader geographical insights, the standardized city-level data are then mapped into three main regions of Vietnam: North, Central, and South. This regional classification facilitates higher-level analysis of recruitment patterns across the country.

The processed location data provide a solid foundation for both city-level and region-level analysis, enhancing the overall understanding of the spatial distribution of accounting and auditing job opportunities.

3.3.7. Textual Data

This step plays a critical role in the data processing pipeline, as it directly affects the accuracy of skill extraction and subsequent analytical results. Job postings typically contain a large volume of unstructured textual information, including job descriptions and requirement sections, which must be systematically processed to extract meaningful insights.

To begin with, the textual data from the job description and requirement sections are collected and merged into a single unified text field for each job posting. Since the original data are primarily written in Vietnamese, all text is translated into English to ensure consistency and facilitate downstream analysis, particularly in keyword extraction and text vectorization.

Text normalization is performed to standardize the data. This process includes converting all text to lowercase, removing punctuation, special characters, and unnecessary symbols, as well as eliminating redundant spaces. These steps help ensure uniformity across all textual data.

Stopwords removal is applied to eliminate common and irrelevant words that do not contribute meaningful information to the analysis. This step reduces noise and improves the quality of extracted features.

The cleaned text is transformed into a structured numerical representation using text vectorization techniques. This allows the model to systematically analyze word frequency and identify important terms within the dataset.

To extract relevant skills, a filtering process is applied based on a predefined dictionary of accounting and auditing-related keywords. This skill dictionary includes a curated list of domain-specific competencies commonly required in accounting and auditing positions. The vectorized text is matched against this dictionary to identify and extract relevant skill keywords from each job posting.

The extracted skills are then standardized and renamed in English to ensure consistency across the dataset. Furthermore, these skills are categorized into three main groups: hard, software, and soft skills. This classification enables more structured analysis of skill demand in the labor market.

The processed textual data provide the foundation for further analyses, including skill frequency analysis and word cloud visualization, which highlight the most in-demand competencies in the recruitment market.

Additionally, the data was prepared for cluster analysis.

3.4. Data Storage

The processed dataset is stored in a structured format to support efficient analysis and visualization. After data cleaning and transformation, the final dataset is saved in CSV format for ease of use and compatibility.

The data are managed using Pandas DataFrames, enabling flexible manipulation and integration across different stages of the analysis. In addition, intermediate datasets may be stored to ensure data integrity and reproducibility of the data processing pipeline.

3.5. Exploratory Data Analysis (EDA)

3.5.1. Dataset Overview

After completing the data preprocessing steps, the final dataset contains attributes and variables related to accounting and auditing, specifically 873 observations and 47 variables. Below are the specific information tables.

Table 3.3. Variable Description

Variable	Data Type	Description
Job Attributes		
job_id	object	Unique identifier for each job posting
experience_min	int64	Minimum years of experience
experience_max	int64	Maximum years of experience
salary_min	int64	Minimum salary offered
salary_max	int64	Maximum salary offered
salary_avg	int64	Average salary calculated from salary_min and salary_max
experience_avg	float64	Average required experience
job_level	object	Job seniority level
Time Variables		
year	int64	Year of job posting
month	int64	Month of job posting
quarter	int64	Quarter of the year
day	int64	Day of posting
Location Variables		
location_list	object	Job location (city/province)
region	object	Region classification

Hard Skills		
accounting	int64	General accounting skills required
reporting	int64	Reporting skill required
finance	int64	Finance-related knowledge
management	int64	Management skill required
monitoring	int64	Monitoring and supervising ability
inspection	int64	Inspection or auditing-related tasks
english	int64	English proficiency required
accounts payable	int64	Accounts payable handling
reconciliation	int64	Account reconciliation tasks
audit	int64	Auditing-related tasks
inventory checking	int64	Inventory checking responsibility
tax finalization	int64	Tax finalization tasks
cash inflow/outflow	int64	Cash flow management
analysis	int64	Data/financial analysis tasks
Software & Technical Skills		
accounting software	int64	General accounting software knowledge
excel	int64	Microsoft Excel skill required
misa	int64	MISA accounting software skill
fast	int64	FAST accounting software skill
sap	int64	SAP system knowledge

erp	int64	ERP system experience
Soft Skills		
communication	int64	Communication skills required
teamwork	int64	Ability to work in teams
honesty	int64	Honesty/integrity requirement
careful	int64	Carefulness/accuracy
responsibility	int64	Sense of responsibility
proactiveness	int64	Proactive working attitude
work under pressure	int64	Ability to work under pressure
detail oriented	int64	Attention to detail
Aggregated Variables		
num_hard	int64	Total number of hard skills required
num_software	int64	Total number of software skills required
num_soft	int64	Total number of soft skills required

Table 3.4 shows details of the statistical descriptions of the study variables.

Table 3.4. Descriptive Statistics

Variable	Count	Mean	Std. dev	Min	25%	50%	75%	Max
experience_min	873	2.107	1.620	0.000	1.000	2.000	3.000	24.000
experience_max	873	4.061	2.572	0.000	2.000	3.000	5.000	29.000
salary_min	873	14.433	6.939	0.000	10.000	13.000	18.000	70.000
salary_max	873	19.687	9.707	500.000	14.000	17.000	25.000	80.000

salary_avg	873	17.060	8.180	450.000	11.500	15.000	20.000	72.500
experience_avg	873	3.084	2.010	0.000	1.500	2.500	4.000	26.500

**All numerical values are rounded to three decimal places.*

**The unit for salary values is millions of VND.*

3.5.2. Visualization

3.5.2.1. Distribution of Salary

The following are figures of the categorical variables of the dataset.

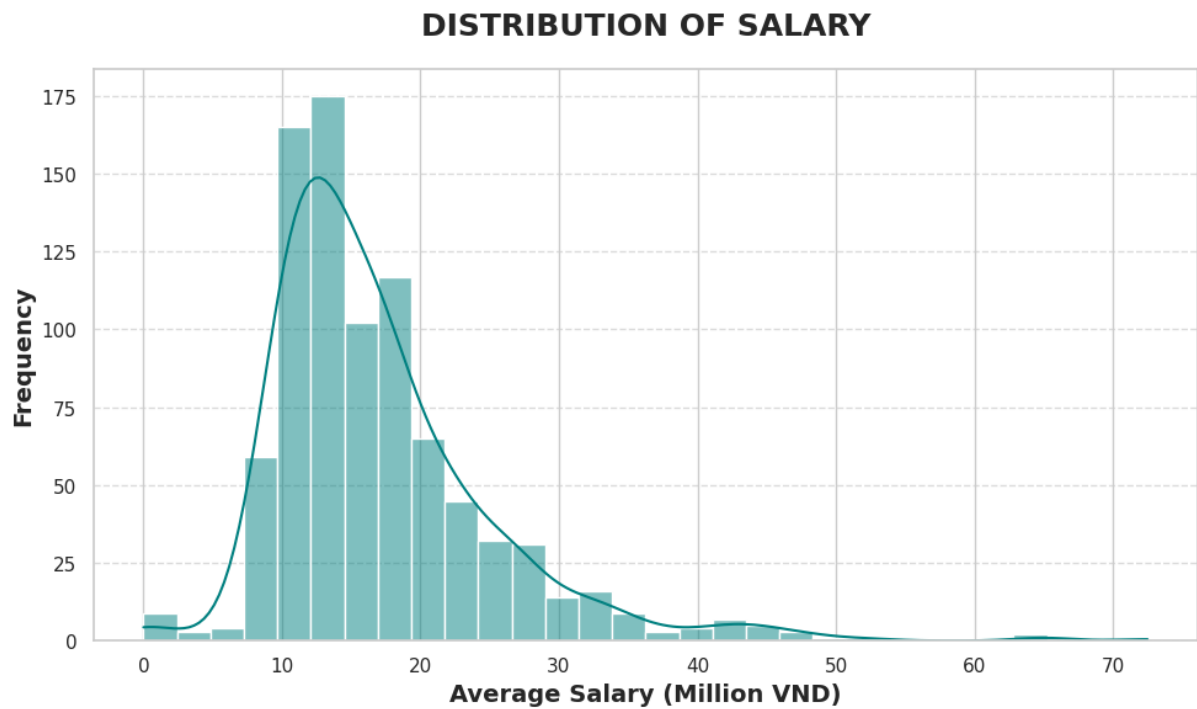


Figure 3.3. Distribution of Average Salary

It's easy to see that the average salary distribution is right-skewed, with most values concentrated in the 10-20 million VND range. The long tail to the right indicates that higher salaries still exist, but they are rare.

BOXPLOT OF AVERAGE SALARY



Figure 3.4. Boxplot of Average Salary

In the boxplot, the median is around 14-15 million VND, which is the market's average salary. The quartile range is quite wide (approximately 10 to 20 million VND), indicating relatively high volatility. Many outliers appear on the higher end of the range (30-70 million VND).

From the above findings, we can conclude that the majority of jobs are in the mid-salary range, with many Senior/Junior positions. Salary increases are not linear when moving up to Manager. Salaries exceeding 40 million VND are only sporadically seen due to the high demands for experience, specialized skills, and Manager positions. Median is lower than mean due to a rightward bias, with mean being pulled up by outliers, which does not accurately reflect typical salaries.

3.5.2.2. Correlation between Minimum and Maximum Salary

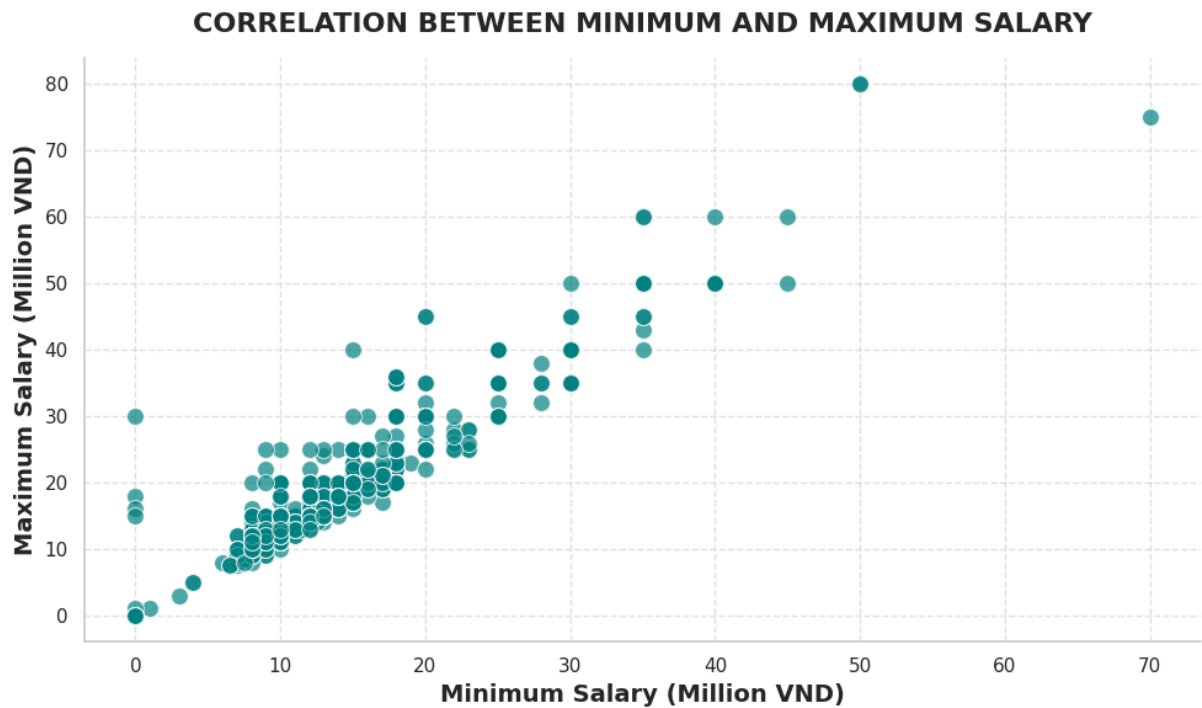


Figure 3.5. Salary Range Relationship

The scatter plot shows a very strong positive correlation between minimum and maximum salaries in job postings. The data is most concentrated in the salary range of 8 to 25 million VND, indicating that this is the most common income range in the market today. The gap between minimum and maximum salaries tends to widen as the base salary increases, reflecting a larger negotiation margin for senior positions compared to entry-level positions with fixed salaries.

3.5.2.3. Distribution of Experience

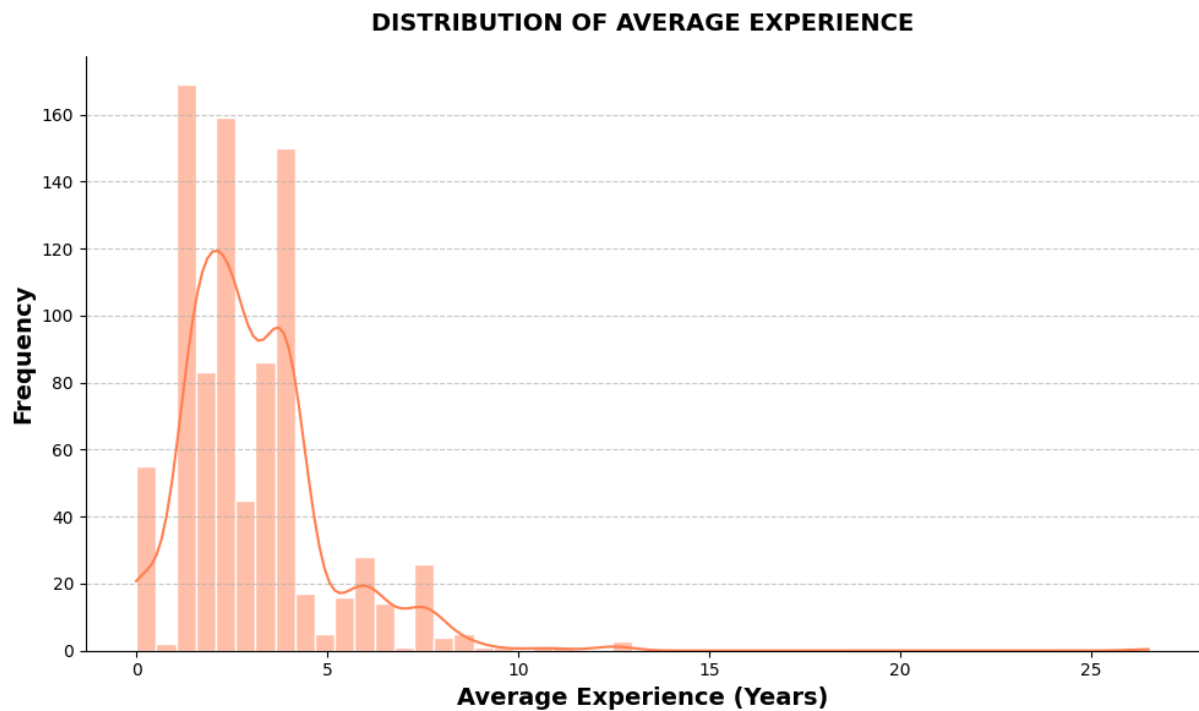


Figure 3.6. Distribution of Average Experience

The experience distribution chart shows that recruitment demand is overwhelmingly concentrated in the young and mid-level personnel group, particularly active in the 1- to 4-year experience range. The density curve (KDE) tends to be distinctly skewed to the right, confirming that the market prioritizes professional execution positions over seasoned management positions. Conversely, the frequency of requests for more than 10 years of experience is very small, reflecting a typical pyramidal workforce structure with a scarcity of senior specialist positions.

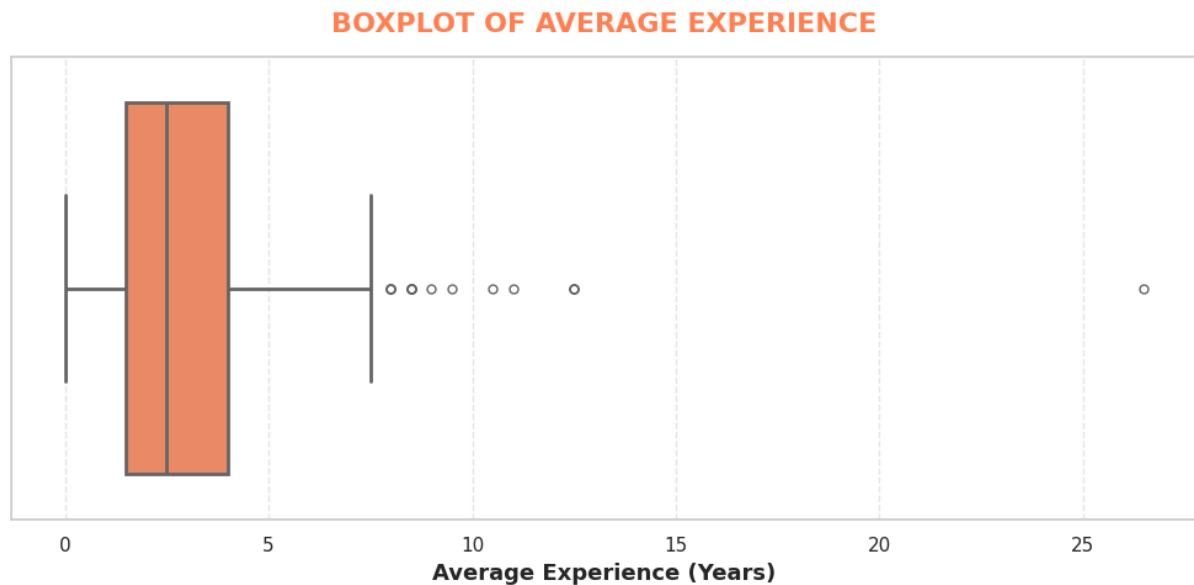


Figure 3.7. Boxplot of Average Experience

The boxplot shows that the majority of market experience demand is concentrated within a narrow range of 1.5 to 4 years, with the median at around 2.5 years. The dense appearance of the circles on the right side of the box's whiskers indicates a large number of outliers ranging from 7.5 years to over 12.5 years, and notably, an extreme point of over 25 years.

This confirms a distinctly right-skewed recruitment market; while the vast majority of opportunities are for young and mid-level personnel, positions requiring high seniority are exceptional cases, accounting for a low proportion and falling outside the typical distribution of the data set.

3.5.2.4. Correlation between Minimum and Maximum Experience

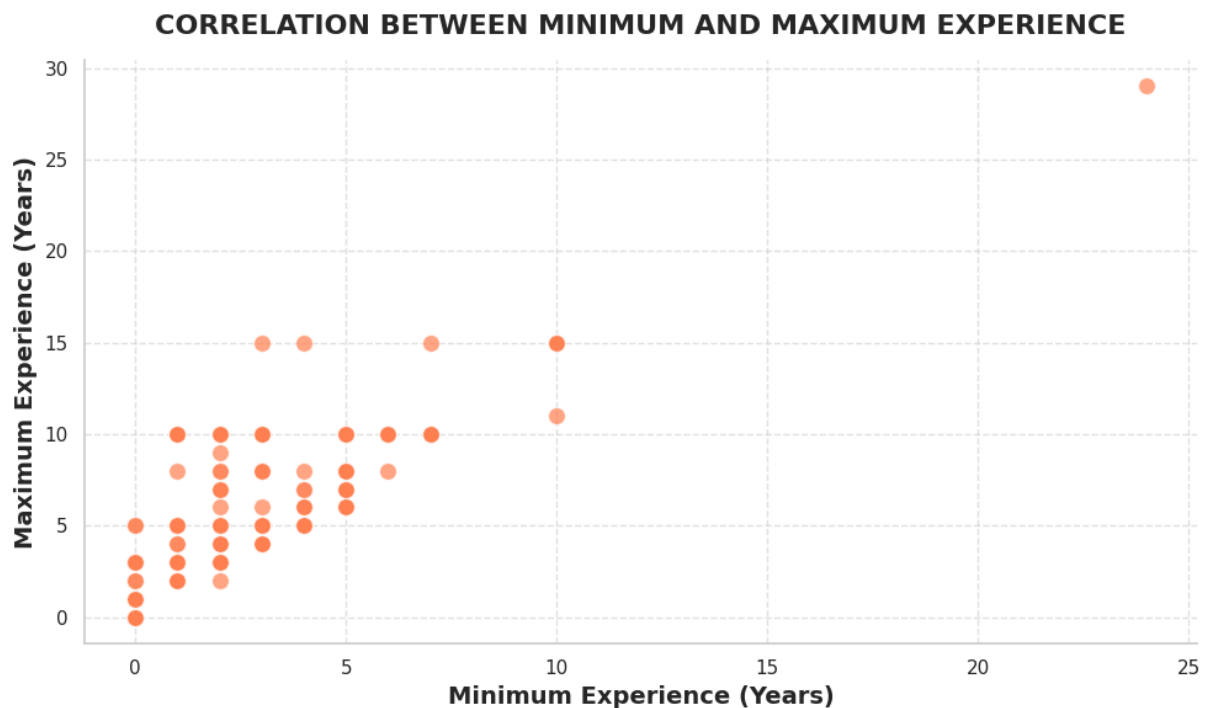


Figure 3.8. Experience Range Relationship

The scatter plot shows a close concentration of experience requirements in the 0- to 5-year range, with the difference between the minimum and maximum typically ranging from 2 to 3 years. Data points thin out as seniority requirements increase beyond 10 years, indicating a more stringent and specific approach to senior-level positions. Notably, the majority of job postings maintain a narrow experience range, reflecting a focus on finding candidates with specific professional skills that are a direct fit for the job, rather than a wide range of experience.

3.5.2.5. Distribution of Job Level

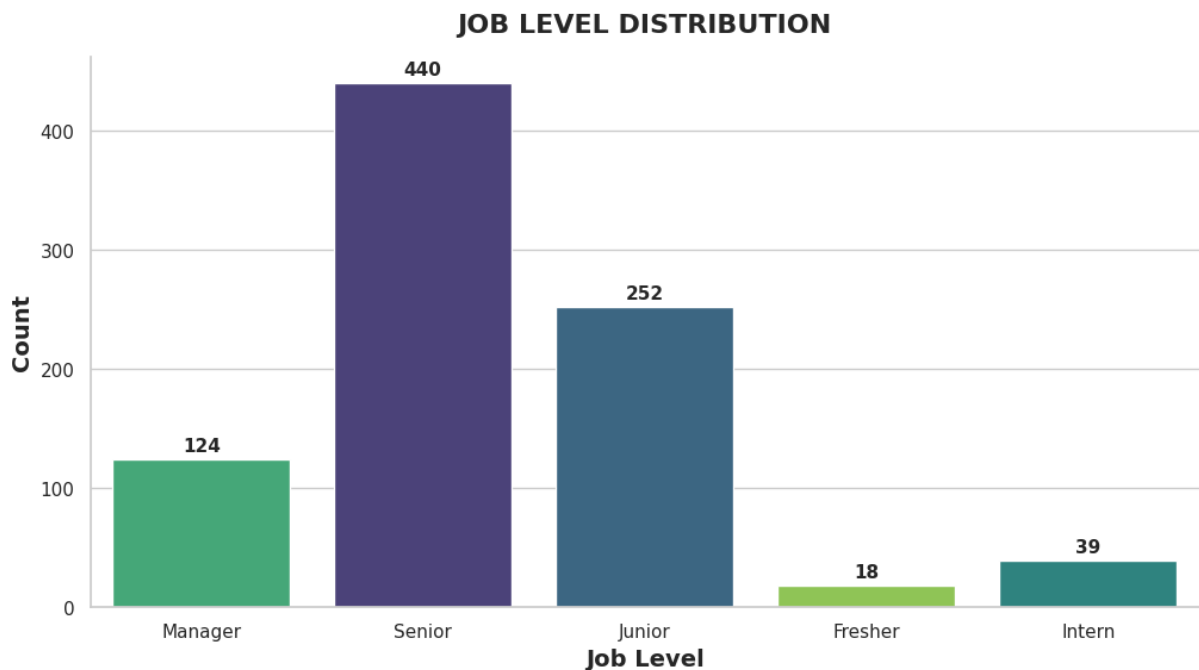


Figure 3.9. Number of Jobs by Job Level

This figure is quite clear that the number of Senior positions is the highest. It has 440 jobs, which is much higher than the others. Junior is the second with 252 jobs, so these two levels take most of the market.

Manager is lower, just 124 jobs, and then Fresher is only 18, and Intern is 39. So the gap is quite big, especially between Senior/Junior and Fresher/Intern. This already shows that there are not many entry-level jobs.

This figure clearly shows that the market is more focused on experienced candidates. If someone is new, it may take more time to find a job, but once they have experience, there will be more opportunities.

3.5.2.6. Distribution of Region

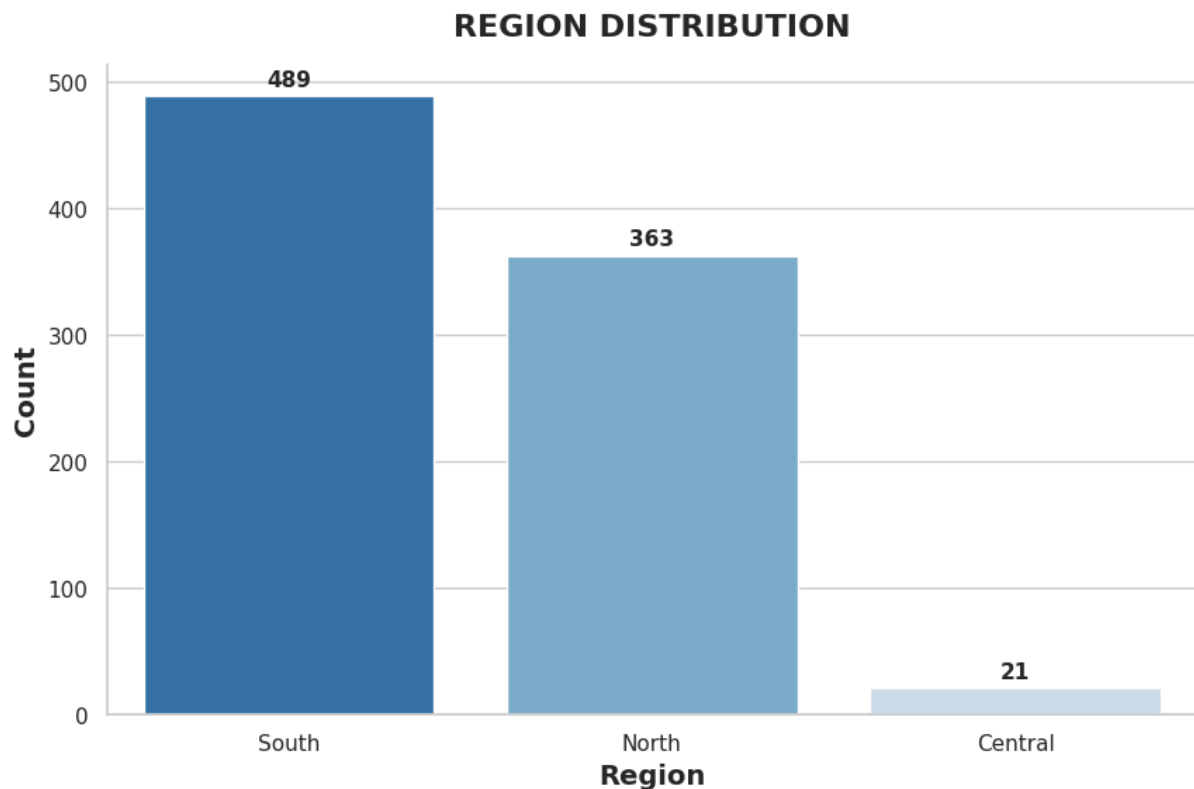


Figure 3.10. Number of Jobs by Region

Figure 3.10 shows that job opportunities in accounting and auditing are mostly concentrated in the South and the North. The South has the highest number with 531 jobs, and the North follows with 406 jobs. Meanwhile, the Central region only has 22 jobs, which is very small and very far from the other two regions. This already makes the difference between regions very clear.

It can also be seen that the job market is not equally distributed in Vietnam. Most of the jobs are still in the main economic regions, especially in the South and the North. This may be because these regions have more companies, more business activities, and also more financial institutions, so they need more accounting and auditing workers. The South is especially strong, maybe because it is one of the most active economic areas in the country.

3.5.2.7. Top 10 Cities by Number of Jobs

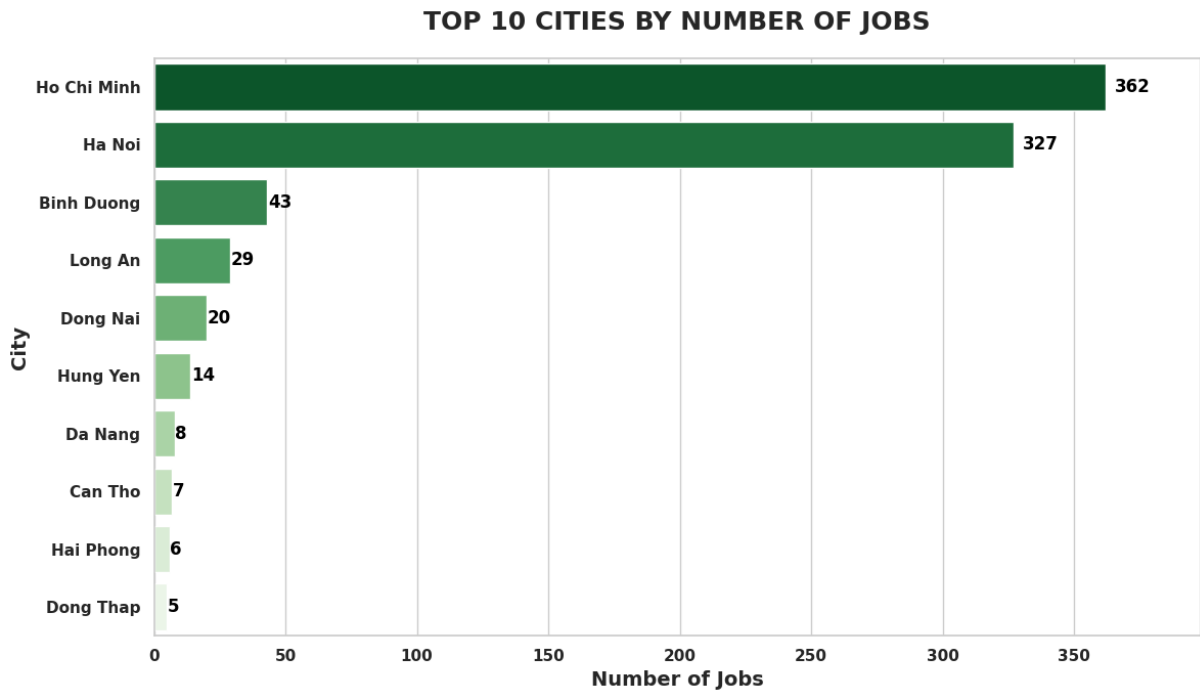


Figure 3.11. Top 10 Cities with the Most Job Postings

Figure 3.11 shows that most jobs are in Ho Chi Minh City (362) and Ha Noi (327). These two are much higher than the others, so it is very clear that they are the main places for accounting and auditing jobs. The difference is really big. For example, Binh Duong only has 43 jobs, Long An has 29, and Dong Nai has 20. These numbers are much smaller compared to Ho Chi Minh City and Ha Noi. So the gap between the top two cities and the rest is very large, not just a small difference.

This shows that job demand is not equal across regions. It is more in places with a strong economy, many companies, and many financial activities. That is why Ho Chi Minh City and Ha Noi have more jobs. Other provinces also have jobs, but not many.

It also means that if someone wants more job chances, they will likely go to these big cities. The market there is bigger and more active, but also more competitive. So overall, this figure clearly shows that most opportunities are still in Ho Chi Minh City and Ha Noi, and other places are much lower.

3.5.2.8. Distribution of Period (Month/Year)

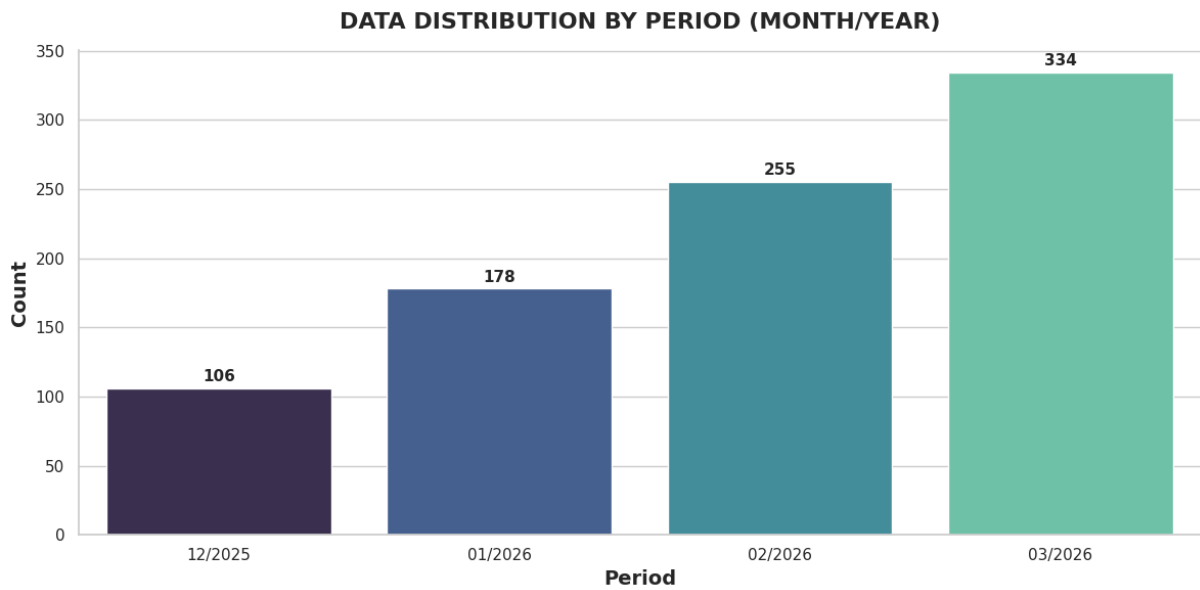


Figure 3.12. Number of Jobs by Period

Statistical analysis of monthly data samples reveals a clear and consistent growth trend over time. Starting in December 2025 with 106 samples, the data size recorded a significant increase in the early months of 2026. Specifically, the number of job postings increased to 178 samples in January 2026, continued to reach 255 samples in February 2026, and peaked at 334 samples in March 2026.

This cumulative increase demonstrates the gradual revitalization of the labor market after the end of the lunar year. In particular, the large amount of data concentrated in February and March (accounting for over 60% of the total sample) indicates that this is the "golden" period of the recruitment season, when businesses finalize their new year's human resources plans and intensify their search for candidates. With a strong focus on data from the first quarter of 2026, the analysis of occupational characteristics and salaries in this study is highly relevant and accurately reflects the current state of the market.

3.5.2.9. Top 10 Most Required Skills

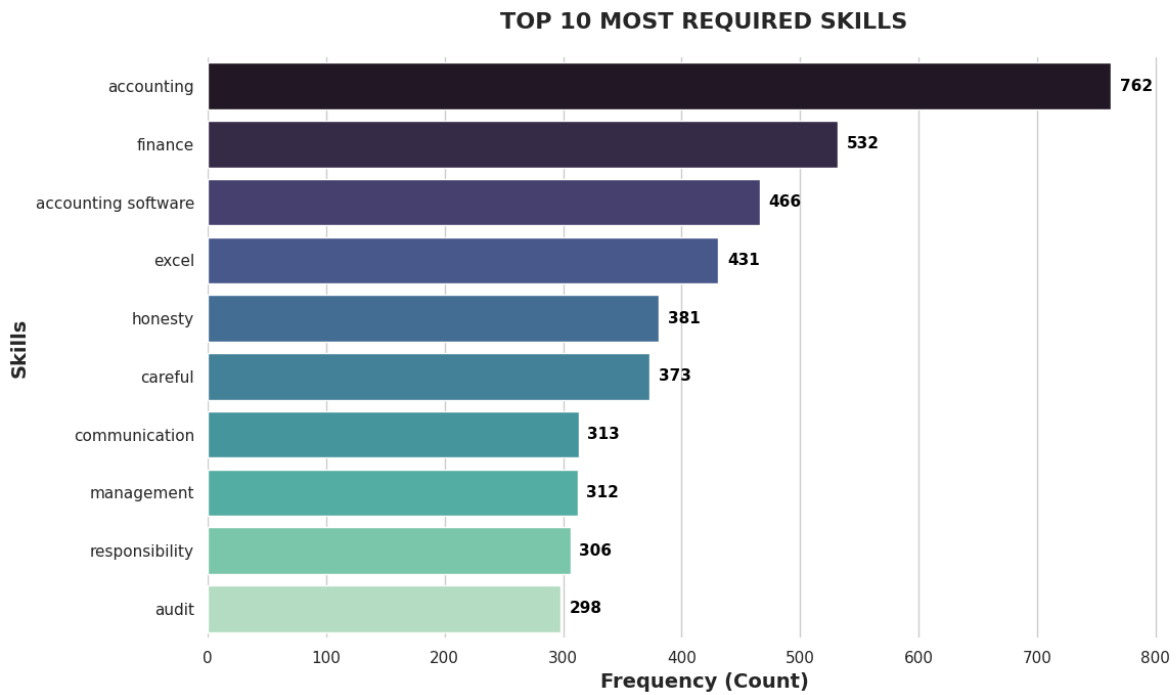


Figure 3.13. Top 10 Most In-Demand Skills

Based on statistical results from job postings, a multi-dimensional competency structure that businesses expect from candidates in the Accounting and Auditing field is evident. First, core professional competencies dominate, with keywords like "Accounting" (762 times) and "Finance" (532 times) appearing frequently. This confirms that, regardless of technological changes or working methods, fundamental professional knowledge remains the most important "passport" and a prerequisite for candidates to enter the labor market.

Alongside professional skills, the ability to apply technology has become a mandatory requirement, rather than just a bonus as before. The appearance of Accounting Software (466 times) and Excel (431 times) at the top of the list shows a strong trend towards digital transformation. Employers today require candidates to not only master accounting theory but also to be proficient in operating specialized software systems (such as MISA, FAST) and spreadsheet data processing tools to optimize work efficiency.

An interesting feature of the data is the emphasis on ethical qualities and work attitudes. Factors such as Honesty (381 times), Careful (373 times), and Responsibility (306 times) ranked very high, even surpassing many other common soft skills. This

reflects the specifics of professions related to finance and data, where integrity and prudence are considered vital values for the business. It shows that employers are willing to prioritize candidates with exemplary attitudes to ensure the safety and transparency of the organization's financial system.

Finally, the balance between management and communication skills (Management and Communication), with frequencies of 312 and 313 respectively, indicates the requirement for a well-rounded employee profile. In the modern context, accountants no longer work in isolation with numbers but need the ability to interact and connect across departments, and even participate in project coordination and management. The combination of these elements creates a new generation of "competency framework," where the integration of professional skills, technology, and ethics creates sustainable competitive value for candidates in the market.

3.5.2.10. Correlation Heatmap

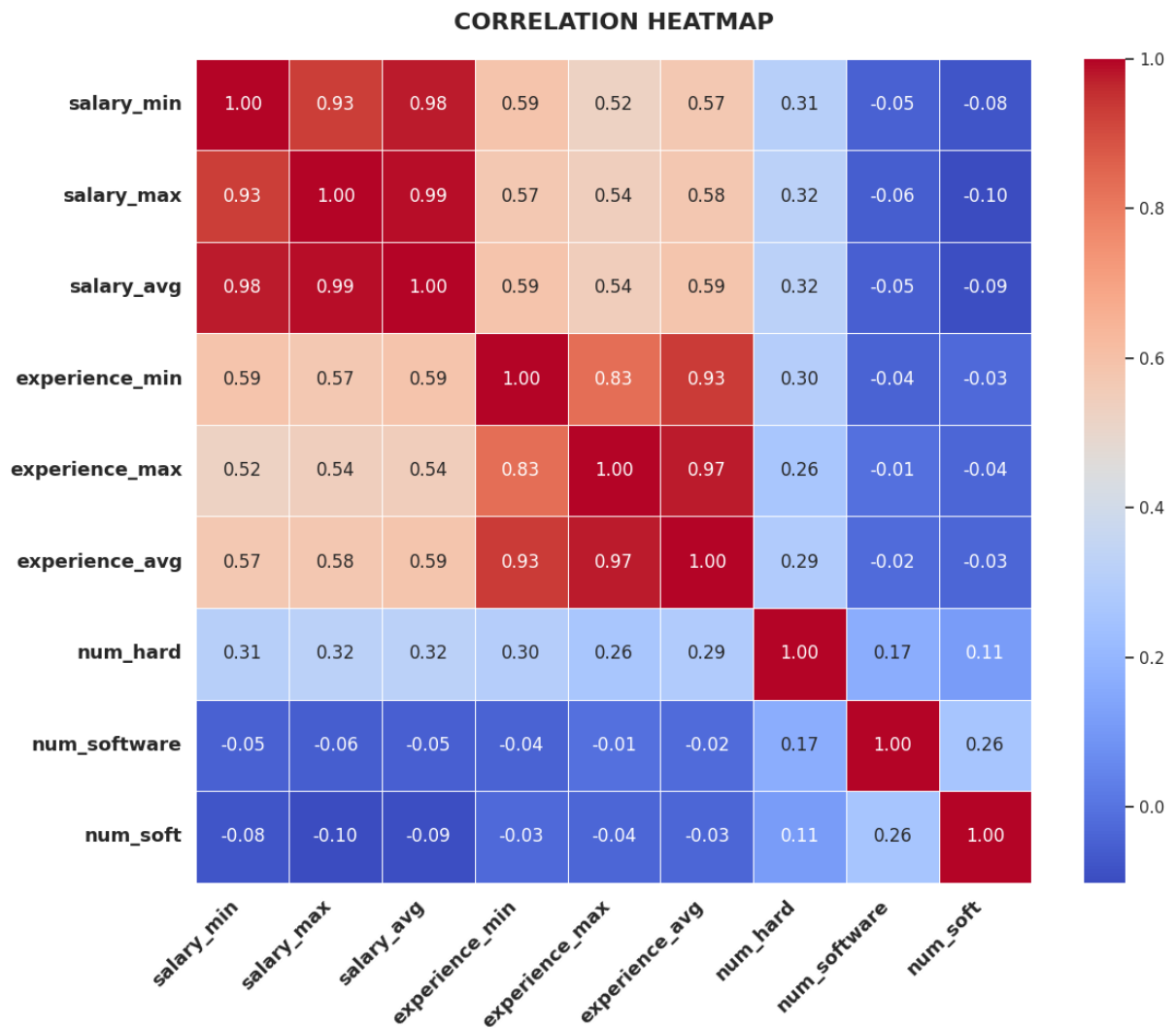


Figure 3.14. Correlation Heatmap of Numerical Variables

The correlation heatmap reflects the multidimensional relationships between key factors in the recruitment data. Most notably, there is a relatively strong positive correlation (approximately 0.54-0.59) between the experience variable and income, confirming that seniority remains a crucial factor determining salary. However, the numbers of soft skills (num_soft) and software (num_software) show a very low, even slightly negative, correlation with salary, suggesting that income depends more on professional quality than on the number of skills.

CHAPTER 4: RESULTS AND DISCUSSION

4.1. Data Analysis

To better understand the relationships and draw conclusions about recruitment demand in the accounting and auditing market, the features are analyzed through visualizations. These visual results provide valuable insights into key patterns and trends.

4.1.1. Distribution of Salary and Experience by Job Level

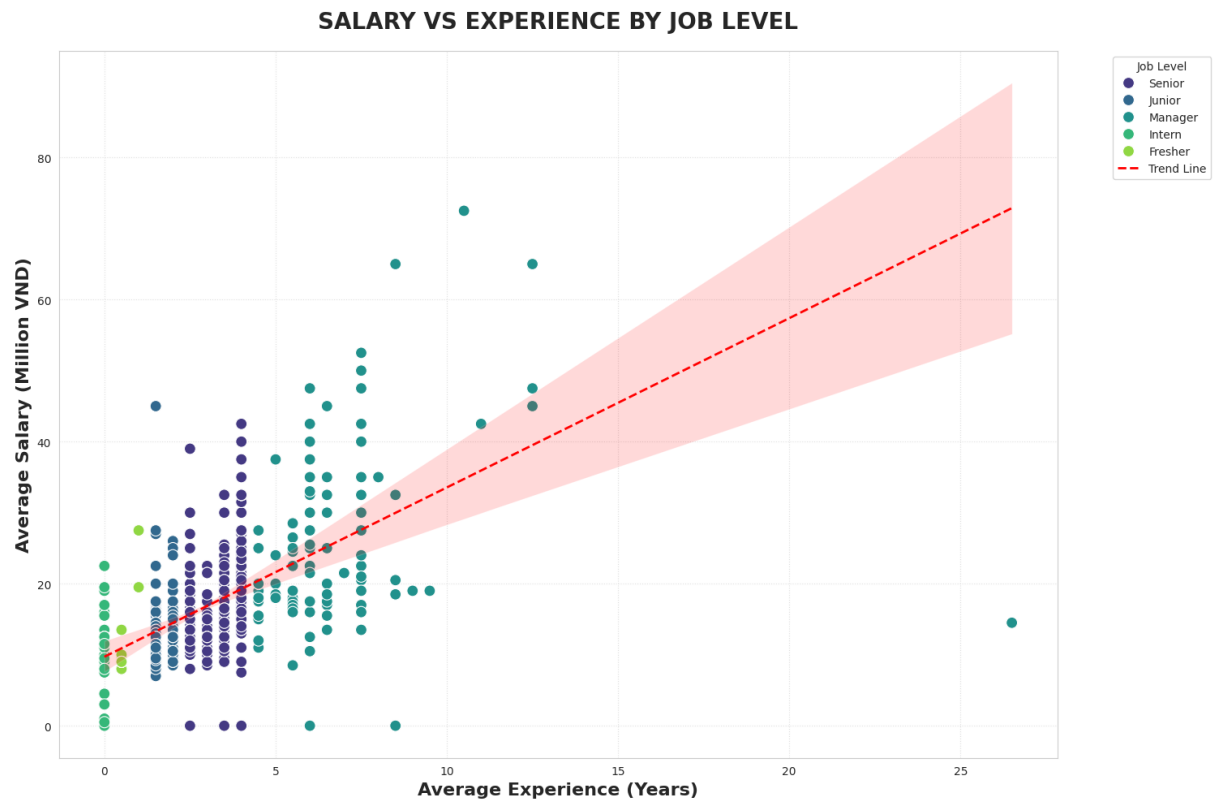


Figure 4.1. Distribution of Salary and Experience by Job Level

This scatter plot shows a clear and positive correlation between years of work experience and financial compensation. The dashed red trend line serves as a predictive basis, indicating that as an individual accumulates more experience, their earning potential will grow steadily according to statistical data.

In the low-experience group (0 to 3 years), the data points are very densely concentrated, mainly representing Intern, Fresher, and Junior positions. During this period, salaries are quite narrow, typically ranging from 5 million to 20 million VND. This concentration suggests a highly standardized entry-level labor market where "market prices" are clearly defined, and there is little room for negotiation until candidates demonstrate certain professional competence.

When experience exceeds 5 years, a significant structural shift begins to emerge. The data points for Senior and Manager levels (green and blue) start to spread more widely vertically. While many Managers earn between 30 million and 50 million VND, the widening of the pink confidence interval suggests that years of experience alone no longer guarantee a specific salary. This divergence implies that at senior levels, compensation is more influenced by "soft" factors and advanced technical expertise—such as possessing international certifications (ACCA, CPA) or understanding complex ERP systems like SAP—rather than simply seniority.

The emergence of outliers—such as a Manager position earning nearly 70 million VND with around 12 years of experience, or some highly experienced positions with low salaries—shows the non-linear nature of career advancement in the later stages of a career. These outliers represent specialized niche markets or high-level advisory roles in FDI companies, where the strategic value of personnel far exceeds the industry average.

The chart confirms that while the accounting and auditing industry has a stable and predictable growth trajectory, achieving high incomes requires a shift from data aggregation to strategic analysis at higher levels.

4.1.2. Distribution of Average Salary by Job Level

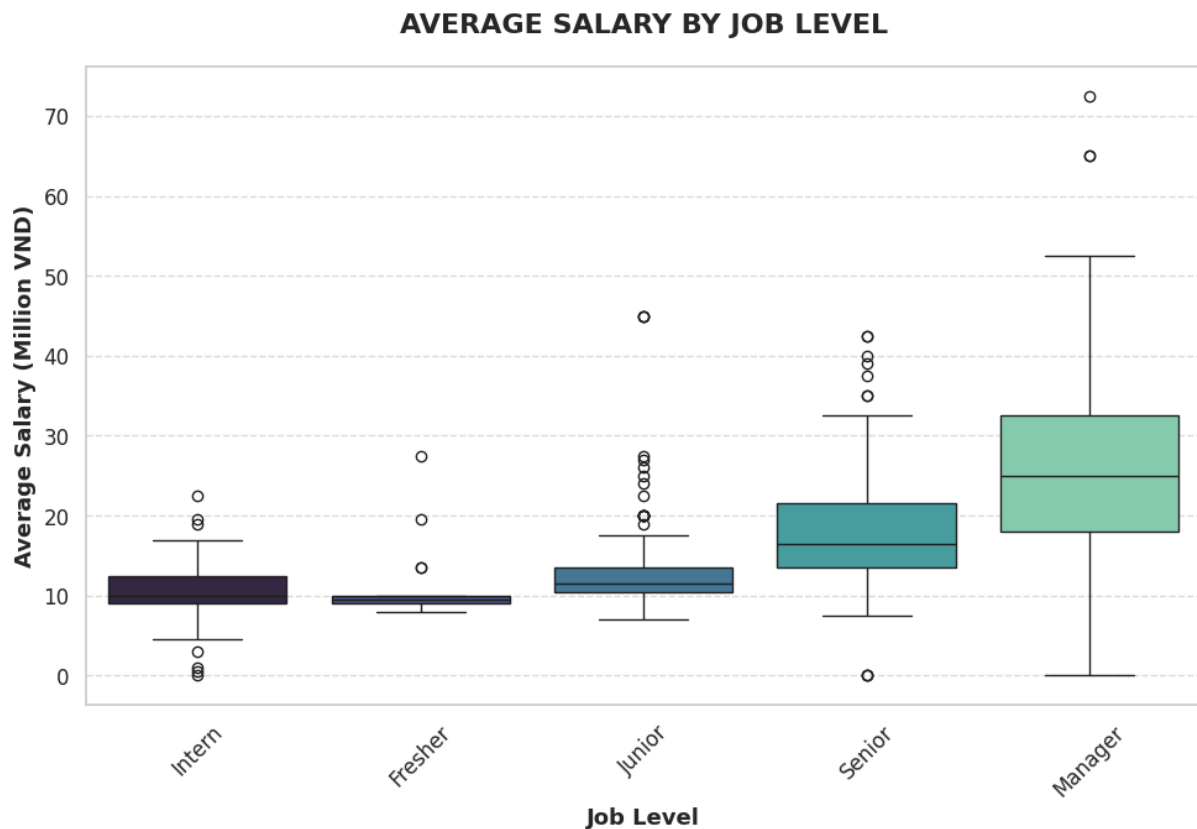


Figure 4.2. Distribution of Average Salary by Job Level

The boxplot shows a pronounced upward trend in monthly compensation, along with an increasing variance in salary ranges, as professionals progress through various hierarchical levels within the industry. This favorable correlation between seniority and financial remuneration accentuates the market's appraisal of accumulated experience, with higher tiers offering not only higher baseline pay but significantly higher earning ceilings.

At the foundational tier, Interns and Freshers exhibit comparable median salaries (historically averaging 10.26 million VND and 11.31 million VND, respectively). The boxplot reveals a particularly tight interquartile range for Freshers, implying that entry-level positions frequently entail highly standardized compensation based predominantly on essential support and educational growth. As individuals advance into the Junior category, the median salary experiences a modest elevation to approximately 12.78 million VND. Notably, the Junior level introduces a wider spread of salaries and several

high-end outliers, indicative of the attainment of independent execution capabilities and varying degrees of technical acumen.

The most pronounced alterations in both compensation and pay dispersion manifest at the elevated strata of the career hierarchy. A significant advancement is noted at the Senior level, where the average remuneration escalates to 17.79 million VND—a surge of nearly 40% when contrasted with Junior positions. The expanded "box" (interquartile range) at this level signifies a broader diversity in pay and a substantial premium attributed to specialized knowledge and the capacity to manage intricate, unsupervised projects.

The trajectory culminates at the Manager level, which commands the apex average salary of 26.13 million VND. This zenith represents an almost 150% enhancement from the Intern tier. Furthermore, the Manager boxplot displays the widest salary distribution and the most extreme outliers (reaching over 70 million VND), underscoring that leadership, strategic decision-making, and high-level accountability serve as the principal catalysts for maximum, highly variable earning potential within the contemporary labor market.

In conclusion, the data elucidate a systematic and meritocratic compensation framework wherein financial rewards—and the potential for exceptional compensation—are significantly calibrated in accordance with professional maturation and managerial influence.

4.1.3. Distribution of Average Salary by Region

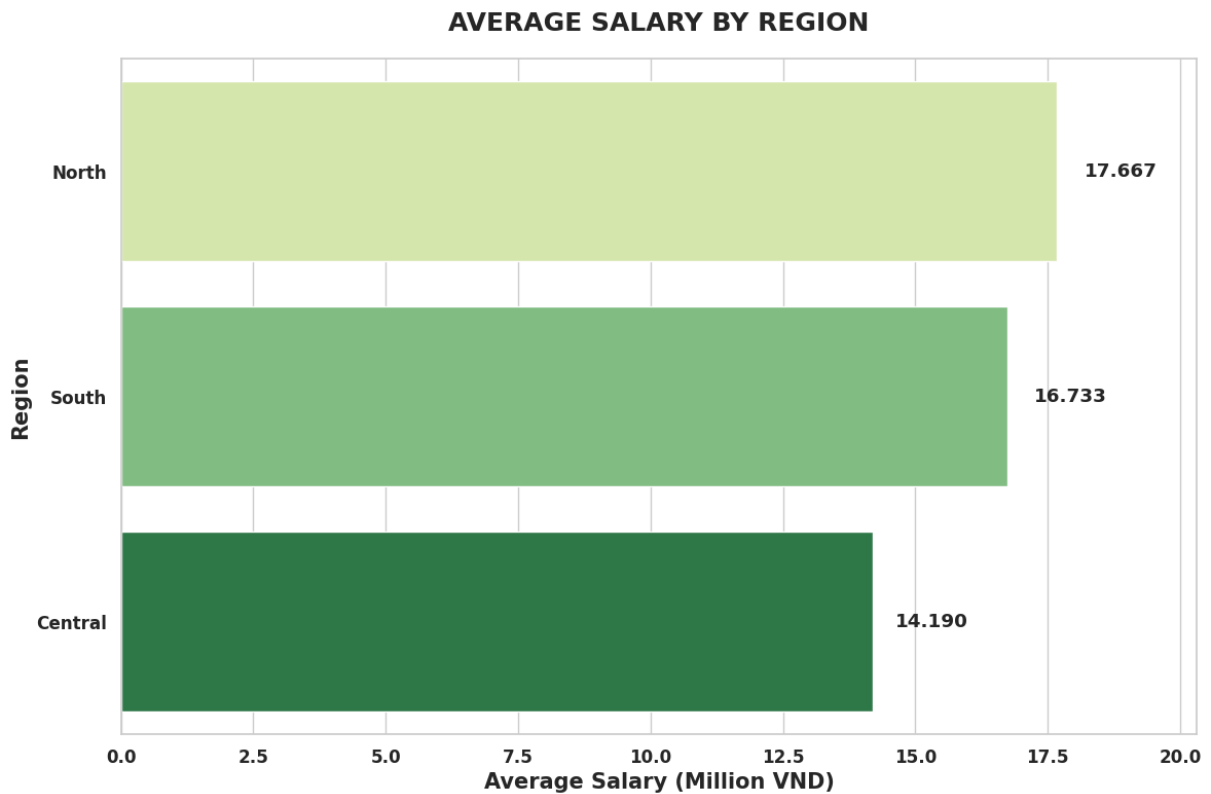


Figure 4.3. Distribution of Average Salary by Region

The chart comparing regional salaries in Vietnam's accounting and auditing sector reveals notable differences in average pay across regions. The data indicates that the North outperforms other regions, with an average salary of approximately VND 17.667 million, higher than the South's average of VND 16.733 million. When considering the overall labor market size, a contrast emerges: although the South consistently records the highest number of job postings and serves as the country's main economic driver, the North's higher average salary reflects a greater concentration of managerial and senior-level positions that require specialized expertise.

The Central region recorded the lowest average salary at VND 14.190 million, over 20% lower than the North. This gap suggests that the Central market is still largely dominated by traditional-economy roles, whereas both the North and South have increasingly moved into high-value service sectors.

This raises an important question: should regions prioritize increasing job quantity or enhancing job value? While the Southern market excels in creating employment opportunities, the Northern economy currently offers the most attractive

financial returns on average. These differences underscore the need for each region to tailor its development strategy according to its strengths and needs.

4.1.4. Distribution of Average Salary by English Requirement

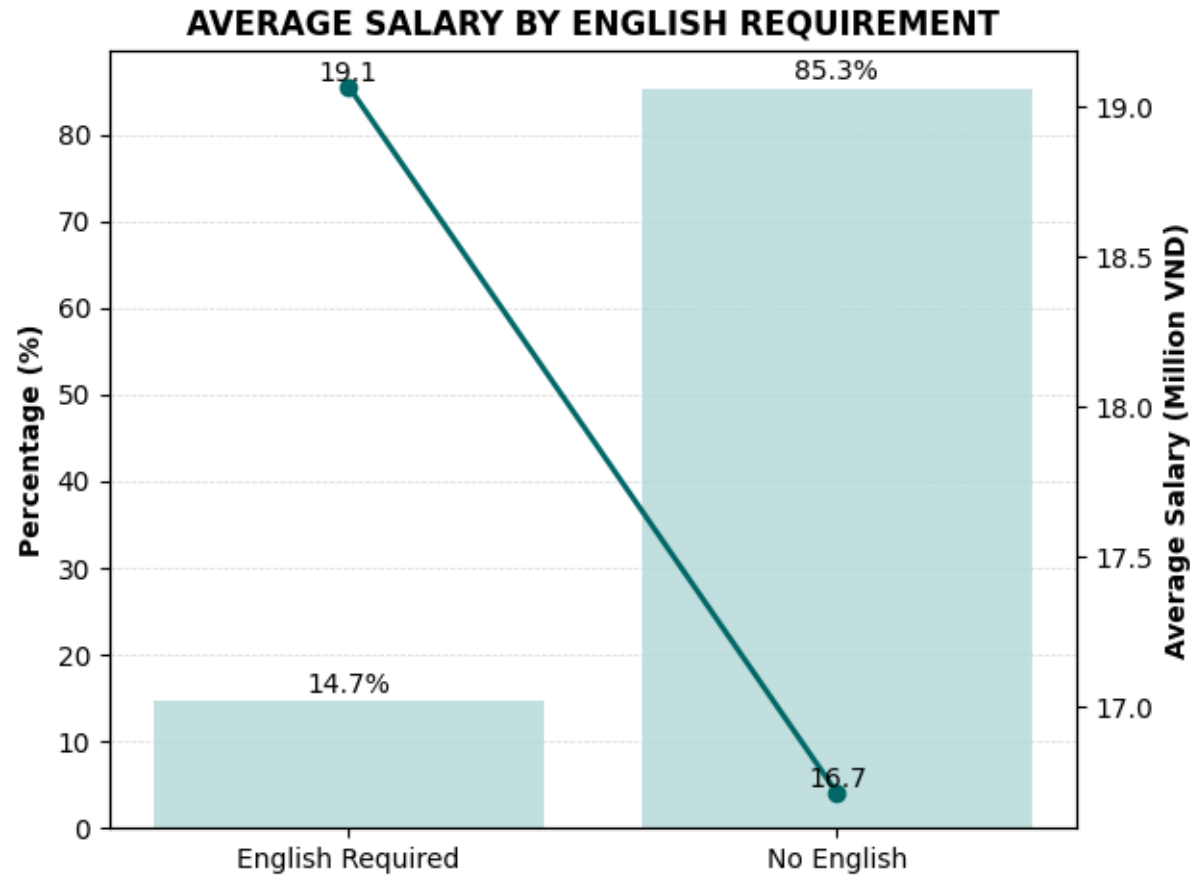


Figure 4.4. Impact of English Requirement on Average Salary

The figure above shows the correlation between English proficiency requirements and two key indicators: job allocation rate and average monthly salary (in millions of VND).

There is a significant difference in the number of jobs between the two groups, with the vast majority (85.3%) not requiring English, while only a small fraction (14.7%) specifically requires English in their job descriptions. Despite being much smaller in number, positions requiring English have a significantly higher average salary, at 19.1 million VND. Conversely, jobs not requiring English have an average salary of only 16.7 million VND.

On average, candidates applying for positions requiring English earn 2.4 million VND more per month than the rest of the group, equivalent to a salary increase of approximately 14.4%. Although the dataset shows that the job market is dominated by

positions not requiring English, there is clearly a strong financial incentive to acquire this language. English is now very common globally as well as in Vietnam, serving as a communication tool that helps people access better-paying opportunities.

4.1.5. Distribution of Average Salary by Number of Hard Skills

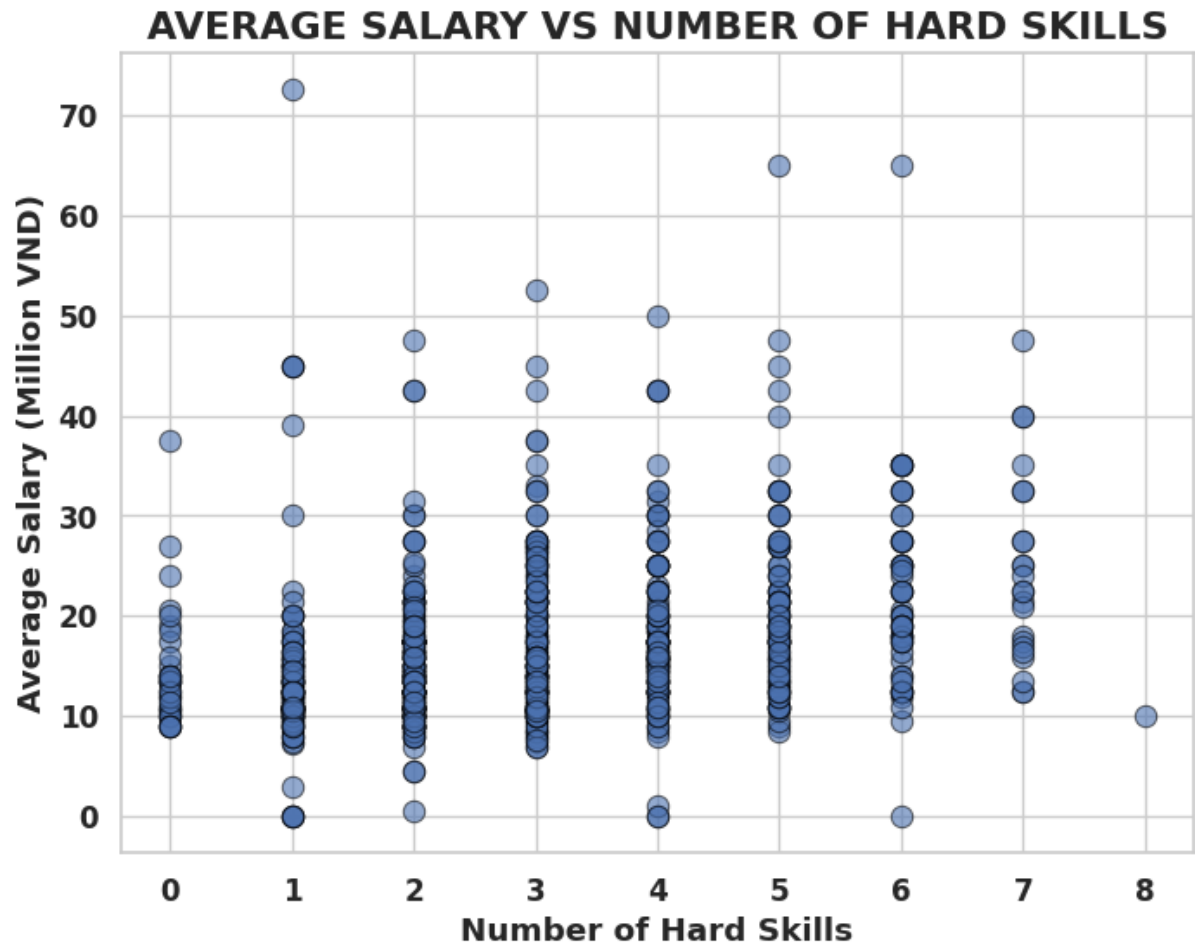


Figure 4.5. Impact of Hard Skills on Salary

Research data indicates that the accounting and auditing employment sector is presently experiencing its most dynamic phase in early 2026, characterized by a pronounced increase in job postings, particularly peaking in March 2026. The ideal candidate profile today is not solely predicated on seniority but is also contingent upon a confluence of numerous factors. Distribution analysis reveals that demand is heavily concentrated within the cohort possessing 2 to 4 years of experience, with prevalent salary ranges spanning from 8 to 25 million VND. Notably, proficiency in foreign languages confers a distinct financial advantage, augmenting average income by approximately 14% in comparison to roles that do not necessitate English language skills.

Interestingly, findings derived from the correlation matrix and scatter plot suggest that the quantity of hard skills alone does not exhibit a direct proportionality to salary. Conversely, income tends to escalate significantly with job level, attaining its zenith at the Manager position with an average of VND 26.13 million. This underscores the reality that the economic value a candidate garners is more contingent upon their profound expertise and managerial competencies than on the sheer number of tools they employ. Ultimately, the emergence of extreme values in salary (up to VND 80 million) and experience (exceeding 25 years) corroborates the existence of an elite segment of personnel, wherein recruitment and compensation standards diverge from the prevailing norms of the mass market.

4.1.6. Distribution of Average Salary by Number of Software Skills

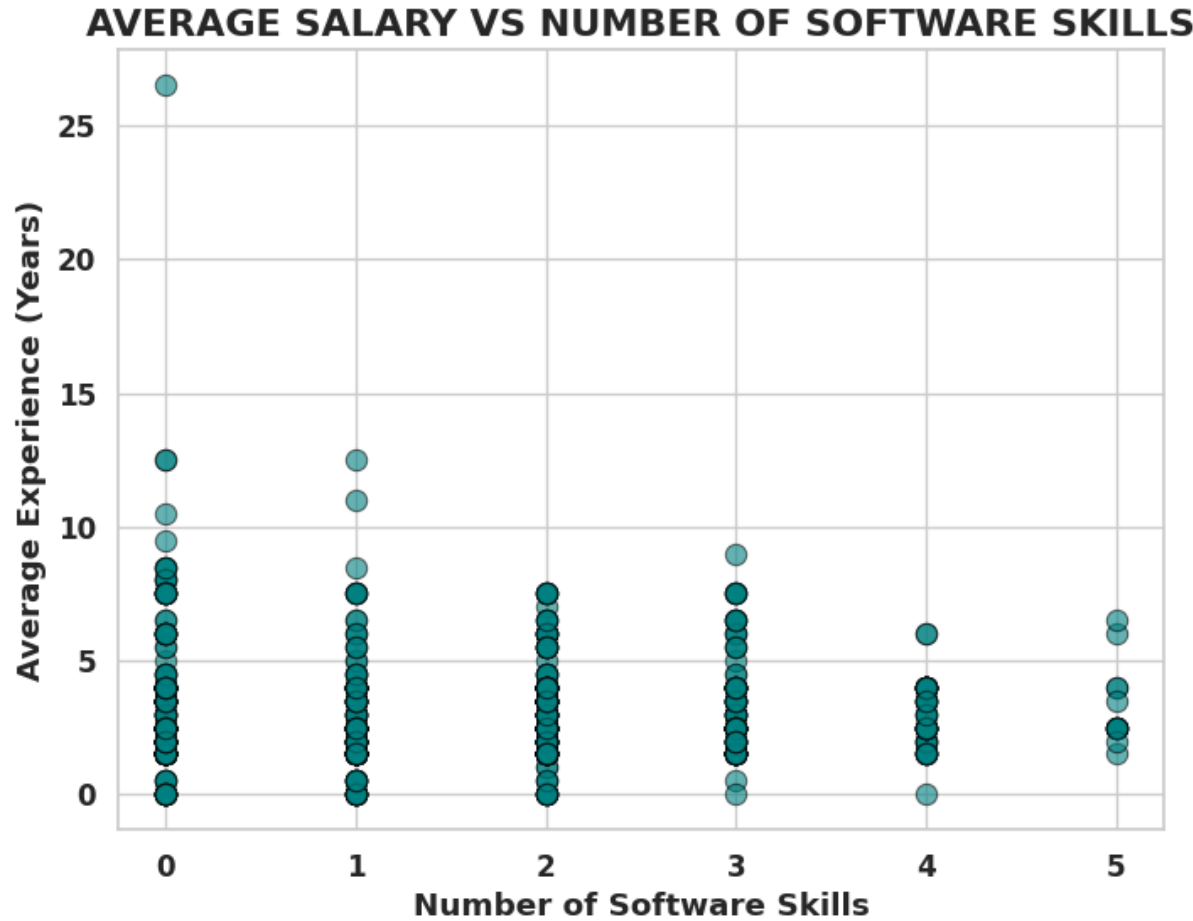


Figure 4.6. Impact of Software Skills on Salary

The study reveals significant findings concerning the impact of skills on income. Data indicates that the quantity of hard and soft skills is not directly linked to salary levels. The correlation coefficients between these variables and average salary are

minimal and somewhat negative. This suggests that the presence of skills is merely a baseline requirement, with true value derived from factors like seniority and management abilities.

A deeper analysis of outliers through scatter plots reveals that high-paying positions (over 60-70 million VND) typically don't require a wide range of skills. Instead, these jobs focus on a group of 1 to 5 specialized skills. This demonstrates that the market prioritizes in-depth expertise and mastery of core tools (such as specialized Accounting Software or advanced Excel) over superficial knowledge of multiple software programs. The data points indicate a predominance of 0-3 software requirements, emphasizing that practical experience and business acumen are fundamental indicators of professional competence.

The research findings validate the presence of a distinct income hierarchy. Although an increase in skills does not significantly enhance salary, job level and language proficiency are the principal determinants of income. Transitioning from a Junior to a Manager role can result in a salary increase of up to 100%, and English proficiency can augment average income by 14%. These elements collectively illustrate a pragmatic labor market landscape: employers prioritize candidates with strategic experience, language proficiency, and expertise in essential business domains over those with merely extensive supplementary skills.

4.1.7. Salary vs. Experience: Insights from K-Means Clustering

To move beyond simple linear correlations, this project utilizes K-Means clustering to categorize the workforce into homogenous groups. This approach allows us to automatically detect natural groupings of employees, providing a deeper understanding of how income levels evolve in relation to professional tenure without the need for pre-defined labels.

4.1.7.1. *Elbow Method*

Before applying the K-Means algorithm, we must determine the ideal number of clusters (k). Using the Elbow Method, which involves running the algorithm for a range of k values and calculating the Within-Cluster Sum of Squares (WCSS).

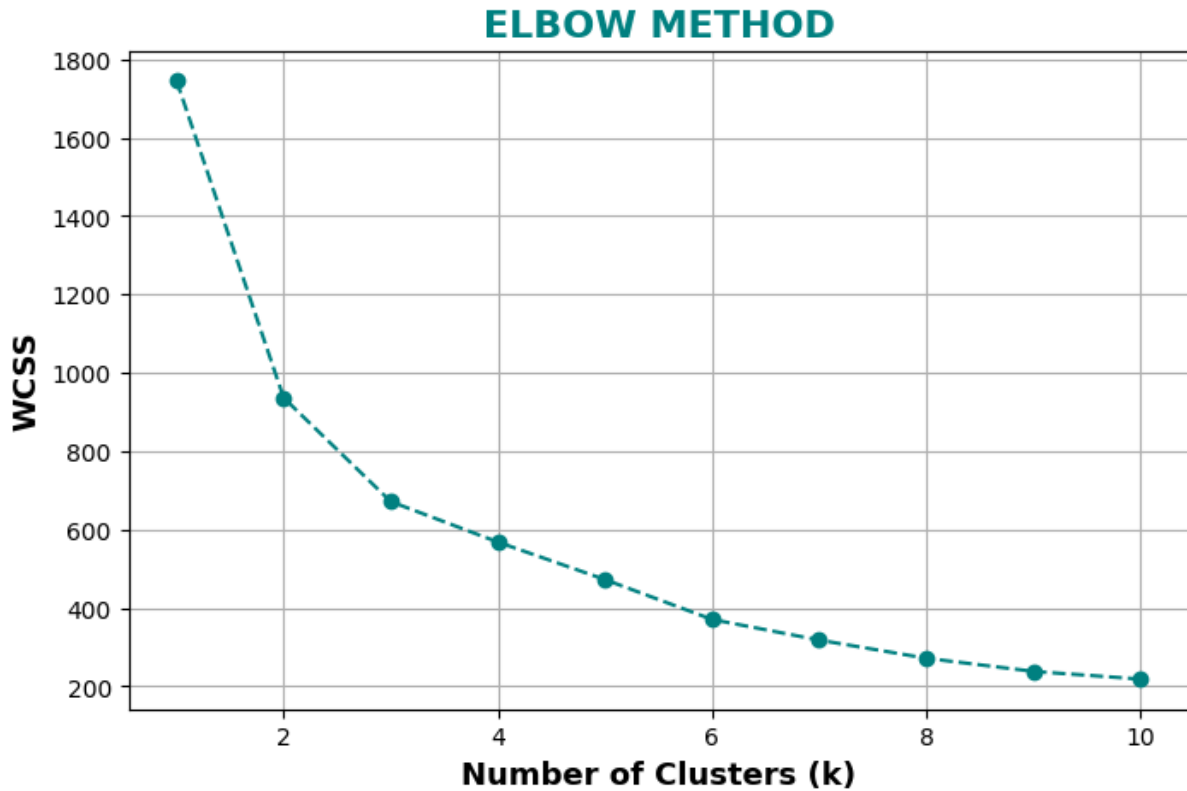


Figure 4.7. Elbow Method

Based on technical analysis from the Elbow chart and empirical results, selecting $k=3$ clusters is considered the optimal decision for classifying accounting market data. The Elbow Method shows that the WCSS (Wave Squared Distances within Clusters) decreases most sharply as the number of clusters increases from 1 to 3, after which the curve begins to flatten. The distinct break point at $k=3$ marks the ideal balance between minimizing model error and maintaining simplicity, avoiding overfitting when attempting to divide the data into too many unnecessary groups.

The three clusters perfectly reflect the natural hierarchical structure of career paths in the Vietnamese accounting industry. Instead of creating overlapping, difficult-to-define groups, $k=3$ has separated the data into three strategic segments: entry-tier, mid-tier, and senior/manager-tier. This division provides a visual overview of "tipping points" in seniority and income, making it easier for businesses to categorize candidates and for employees to clearly identify their position on the professional growth map.

To examine the relationship between work experience and income, the following scatter plot visualizes the distribution of observations across these two variables,

allowing the identification of distinct income groups associated with different experience levels.

4.1.7.2. Distribution of Salary and Experience (K-means)

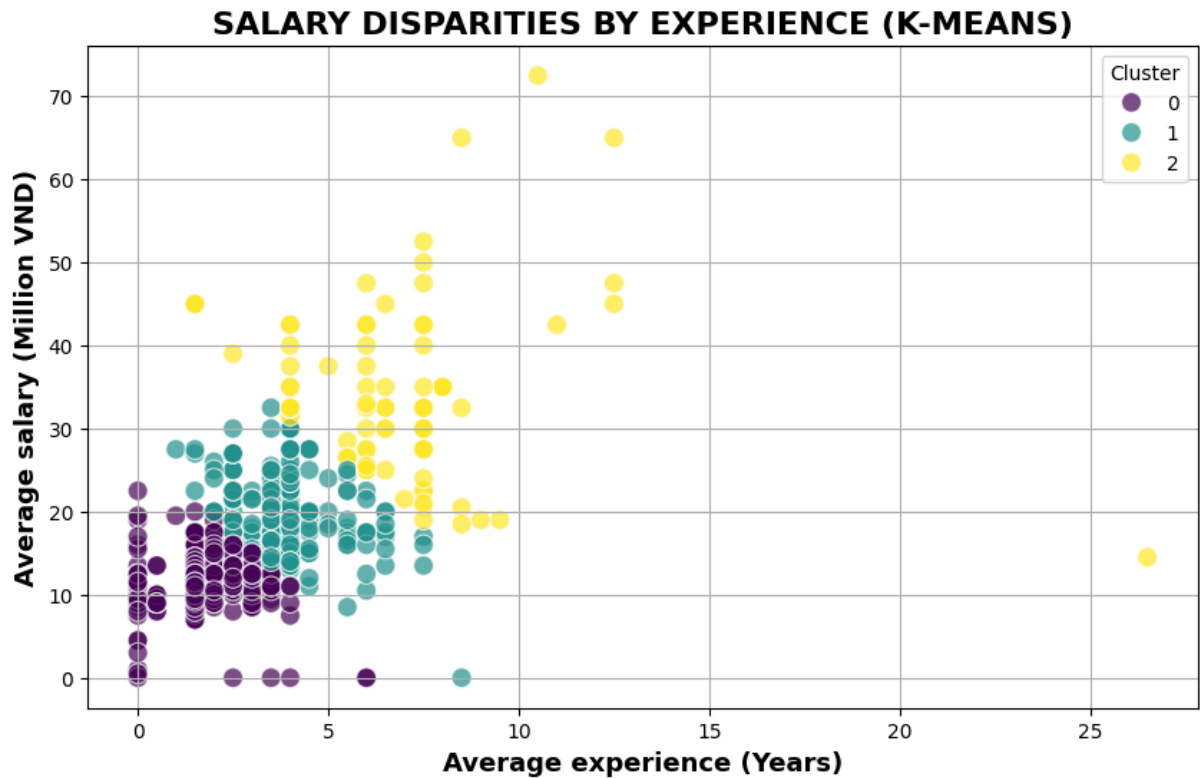


Figure 4.8. Salary Disparities by Experience (K-means)

The K-means clustering model with $k=3$ has divided the accounting labor market into three distinct segments, reflecting the strong correlation between seniority and income. Cluster 0 (purple) represents the "Entry-level" group with less than 4 years of experience and incomes primarily below 20 million VND, representing the highest concentration. The core insight here is that the early career stage exhibits high salary homogeneity, where Intern, Fresher, and Junior candidates are in the process of acquiring foundational skills.

Moving up the ladder, Cluster 1 (green) outlines the "Middle-market" group, comprising personnel with 2 to 8 years of experience and stable salaries ranging from 15 to 35 million VND. A notable point is the overlap between this cluster and the expert group, indicating that for the same number of years of experience, salaries begin to differentiate significantly based on individual capabilities rather than length of service.

This is a crucial transitional phase where differences in specialized skills and professional certifications begin to create a gap in economic value.

Finally, Cluster 2 (yellow) is the "High-value/Specialists" segment for management and professional positions with 5 to over 12 years of experience, with salaries ranging from 20 million to over 70 million VND. This group has the widest dispersion, demonstrating that at the high level, the market no longer has a single price but depends on each individual's ability to solve complex financial problems. Overall, the model shows that the salary axis has a stronger clustering effect than the experience axis, confirming that the market is rewarding actual competence rather than pure seniority.

4.1.7.3. Clustering Jobs

The figure below illustrates the relationship between experience and salary, along with the centroid points representing the average characteristics of each group.



Figure 4.9. Clustering Jobs by Salary and Experience

Based on the cluster analysis chart from the K-means algorithm, we can clearly see the income and seniority development path in the accounting industry, divided into three strategic phases. The first cluster represents the entry-level group, located at approximately 1.9 years of experience with an average salary of approximately 12 million VND. This is the convergence point of the vast majority of new employees, reflecting the fact that in the first two years of their careers, accounting salaries are highly consistent and show little variation among individuals. A key insight at this stage is stability, where the market primarily pays salaries based on the performance of basic tasks rather than differentiation in specialized skills.

Moving into the intermediate stage, the center of the second cluster shifts to 3.8 years of experience, with an average salary approaching 20 million VND. The gap between the first and second clusters shows that after surpassing the 2-year accumulation threshold, employees often experience a significant jump in income, increasing their market value by approximately 60-70%. This demonstrates that after about 4 years of work, accounting personnel begin to master important skills such as using accounting software (MISA, Fast) or advanced Excel skills, thereby creating higher added value for the business.

The final cluster, at the highest level, represents senior management or financial professionals with an average salary of 34 million VND (approximately 6.8 years of experience). This represents the group of senior managers or financial experts, where income no longer increases steadily but begins to experience significant breakthroughs. The salary gap between the second and third clusters is significantly larger than in the previous period, indicating that the market is willing to offer generous compensation to personnel with over 6 years of experience. The core insight is that the career path in the accounting industry in Vietnam is still highly senior-oriented, but the 7-year mark is the "breakthrough point" for personnel to transition from purely professional roles to strategic management roles with superior compensation.

4.1.7.4. Salary vs. Experience Correlation (K-means)

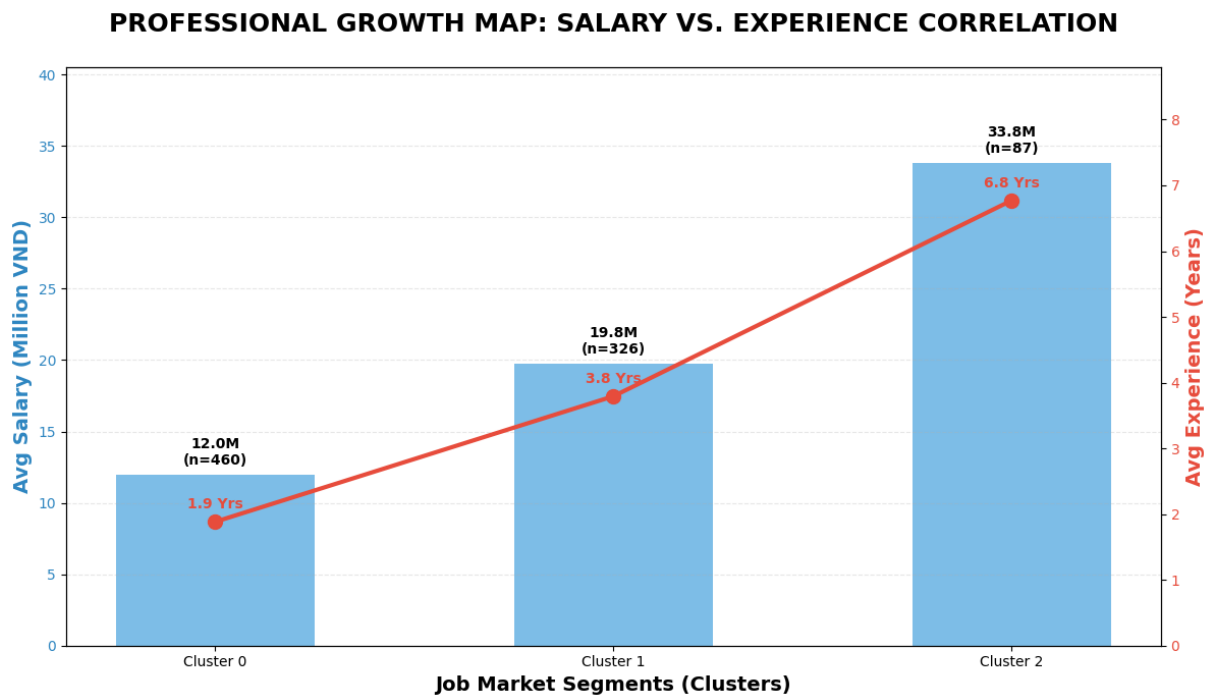


Figure 4.10. Professional Growth Map: Salary vs. Experience Correlation

When applying the K-means clustering algorithm based on salary and experience, the results are divided into 3 groups. The graph shows the relationship between average salary and average experience of each group of workers in the labor market. The results are as follows:

Group 0: 460 job postings, salary ~11.98 million VND, experience ~1.9 years - corresponding to Intern/Fresher positions.

Group 1: 326 job postings, salary ~19.77 million VND, experience ~3.8 years - corresponding to Junior/Senior positions.

Group 2: 87 job postings, salary ~33.78 million VND, experience ~6.8 years - corresponding to Senior/Management positions.

This shows that salary and experience are not directly proportional to workforce size: the smaller group with higher salaries reflects senior positions, while the larger group is younger and more prone to turnover. Group 1 has significantly higher salaries and experience than the others, suggesting that this group needs to focus on employee

retention and advanced training, while Group 2 needs to develop skills to increase salaries and reduce employee turnover.

4.1.8. Job Postings Over Time

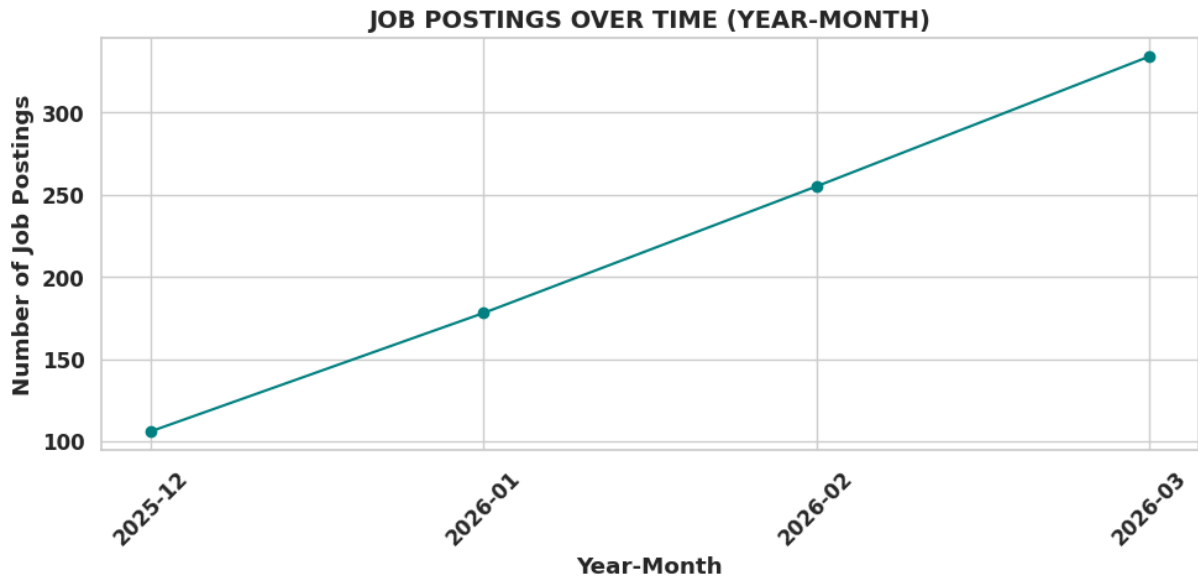


Figure 4.11. Trend of Job Postings Over Time

The chart shows a clear shift in recruitment demand cyclically between the Lunar New Year holiday and the beginning of the fiscal year. At the end of the year (December 2025), the number of job postings was at its lowest (106) due to market stagnation as companies focused on meeting old targets and preparing for Lunar New Year bonuses. However, immediately after the Lunar New Year holiday, the market experienced a strong surge with a continuous upward trend, peaking in March 2026 with 334 postings (more than three times the number at the end of the previous year). This sudden increase not only reflects the early-year personnel turnover but also indicates the urgent need for additional resources by businesses to serve the accounting season and implement new business plans.

4.1.9. Distribution of Region by Job Level



Figure 4.12. Distribution of Region and Job Level

Figure 4.8 shows jobs by region and job level, and it is quite clear that most jobs are in the South and the North. The highest number is Senior in the South with 253 jobs, and in the North, Senior is also high with 178 jobs. So, Senior roles are really the main part of the market, especially in these two regions.

Junior is also high, with 143 jobs in the South and 101 in the North, so not only Senior but also Junior has many opportunities. Manager is lower but still quite visible, like 62 in the South and 59 in the North. So, all these levels are mostly in the South and the North.

The Central region is very low in all levels. For example, Senior only has 9 jobs, Junior 8 jobs, Manager 3 jobs, Fresher 1 job, and Intern even 0. So, it is really very low, almost no jobs compared to the other regions.

Fresher and Intern are low in all regions, like only 6 and 19 in the North, and 1 and 20 in the South. So entry-level jobs are limited everywhere, not just in Central. This again shows that companies prefer people with experience.

This figure shows two things quite clearly. First, jobs are mostly in the South and North, not Central. Second, most jobs are for experienced people, such as Senior and Junior. So, if someone wants more opportunities, they will likely go to the South or North and also need some experience.

4.1.10.Skills Word Cloud



Figure 4.13. Word Cloud Visualization

The illustration presents a keyword cloud highlighting the most in-demand skills in accounting and auditing job postings, where the size of each term reflects its relative frequency. To best visualize skill patterns, they are visualized through a word cloud, just like the data in the figure above. Word clouds express the information more vividly and make comparisons easier without needing to explicitly present numbers.

Looking at the Word Cloud and the "Top 10 Most Required Skills" chart, it is clear that accounting (762 counts) and finance (532 counts) are the most dominant keywords, appearing as the largest terms. This is followed by technical tools like Accounting Software (466 counts) and Excel (431 counts). Interestingly, soft skills and personal attributes such as Honesty (381 counts), Careful (373 counts), and

Communication (313 counts) are also very prominent. This suggests that the market not only values professional knowledge but also places a high priority on the integrity and meticulousness of candidates.

When we link these skills to the "Average Salary by Job Level" and "Boxplot of Average Experience", we see a professional roadmap. The majority of job postings target candidates with 1.5 to 4 years of experience (the interquartile range in the Boxplot), with a median of around 2.5 years. However, the salary increases significantly as one moves up the ladder: from an average of 10.26 million VND for Interns to 17.79 million VND for Seniors, and peaking at 26.13 million VND for Managers. The Word Cloud’s emphasis on management and responsibility aligns perfectly with these higher-paying tiers, where specialized expertise and leadership become the primary drivers of value.

4.1.11. Top 5 Skills Prevalence by Job Level

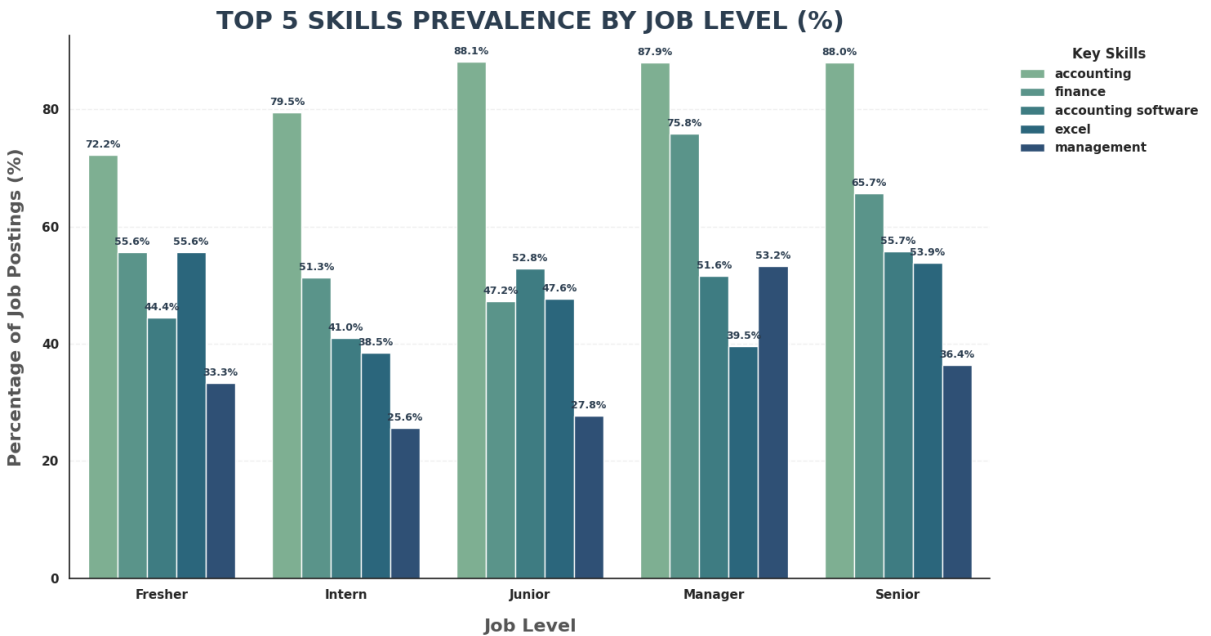


Figure 4.14. Top 5 Skills Prevalence by Job Level

The figure illustrates the prevalence of the top five skills across five job levels.

Accounting remains the dominant skill across all levels, starting at 79.5% for Interns and rising to a high of 88.1% for Junior and 88.0% for th Senior levels. This trend confirms that specialized knowledge is essential for career advancement, and even

entry-level candidates are expected to possess solid professional knowledge from day one.

For Freshers, Accounting Software (44.4%) and Finance (55.6%) are required at a relatively high rate, indicating that this is not simply theoretical work but also requires immediate practical application. The demand for Management skills (33.3%) at this level is notable, as even recent graduates are expected to take responsibility for their assigned duties.

At the Intern level, the demand for Accounting knowledge is 79.5%, which is higher than for Freshers (72.2%). This demonstrates that internship programs are rigorously screening to bring a high-quality workforce to the market. However, the prevalence of Finance (51.3%) and Accounting Software (41.0%) at the Intern level is slightly lower than the requirements for Freshers, as Interns are often provided with a trial period to learn specific tools through on-the-job training.

The Junior level represents the main workforce of the company, where the demand for Accounting skills peaks at 88.1%. Excel (47.6%) and Accounting Software (52.8%) account for a high percentage, with Juniors being the primary users of technical tools to process data in their daily work. This level also serves as a crucial stepping stone, with demand for Management skills at 27.8% as individuals begin to take on roles such as team coordination.

The Manager level represents the most significant transition. While Accounting is still essential (87.9%), notable growth is seen in Management (53.2%) and Finance (75.8%). Managers act as a vital link between executives and technical implementers, requiring a balanced set of leadership and financial expertise.

At the Senior level, Accounting holds the highest percentage (88.0%), followed by Finance (65.7%) and Accounting Software (55.7%), which remain stable and show significant growth. However, the demand for Management skills has decreased to 36.4% compared to the Manager position. This reflects the reality that Senior roles in today's market often function more as expert advisors, handling the most complex and technically challenging models, rather than primarily focusing on team supervision.

4.1.12. Distribution of Skill Groups

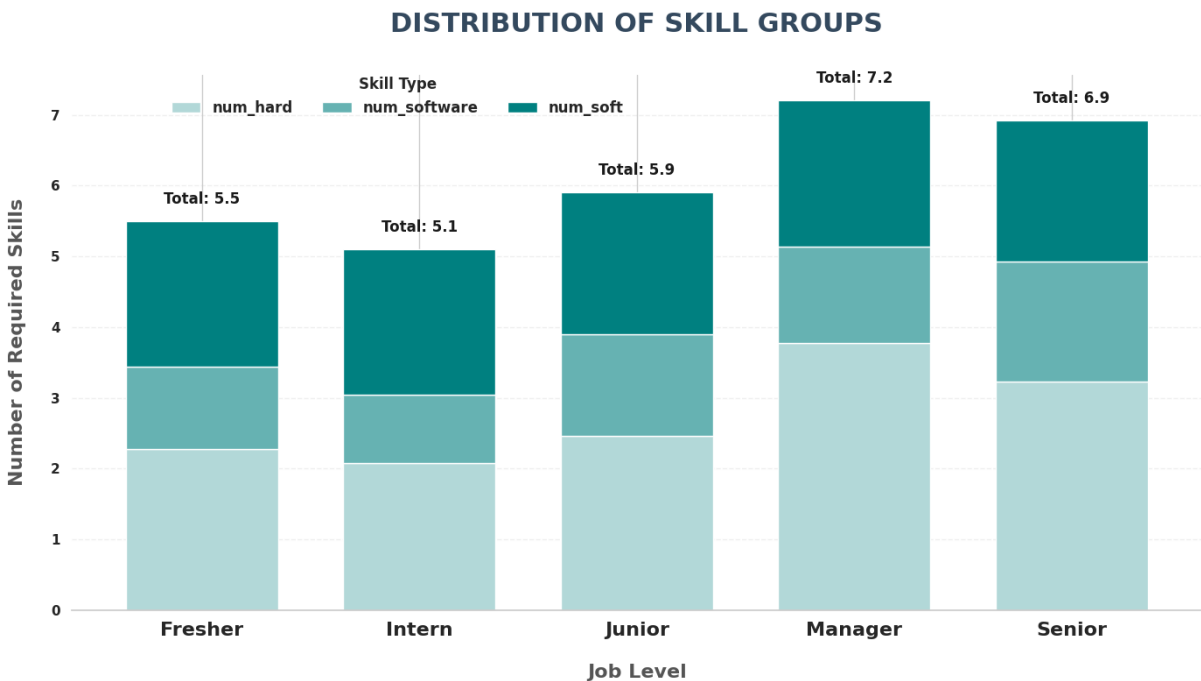


Figure 4.15. Distribution of Skill Groups

This chart illustrates the distribution of skill groups in each job posting at different levels, comprising three skill groups: hard skills, soft skills, and software skills. The results show a significant increase in the total number of skills required; the higher the level, the greater the number of required skills. At lower positions like Intern and Fresher, the total number of required skills is only 5.1 and 5.5, respectively. However, at higher levels like Manager and Senior, this number rises to 7.2 and 6.9. The main difference lies in hard skills; as the level increases, the number of required software and soft skills increases only slightly, while hard skills show more pronounced growth from the entry levels to the Manager level, reflecting the increasing complexity of work and responsibilities.

The chart shows that technical expertise plays a crucial role in promoting career development and advancement. The higher the level, the greater the demands for hard skills—reaching a peak of 3.8 skills at the Manager level—indicating that specialized knowledge is a key factor for progress to higher positions. Software and soft skills also play an important role; the relatively consistent number of required skills across levels (ranging from 1.0 to 1.4 for software skills and 2.0 to 2.1 for soft skills) shows that these

skills are prerequisites and the basic foundation for maintaining and developing a career. Regardless of the position or level, soft and software skills are always essential, highlighting their role in work, collaboration, effective communication, and leadership.

4.1.13. Skills Correlation Matrix

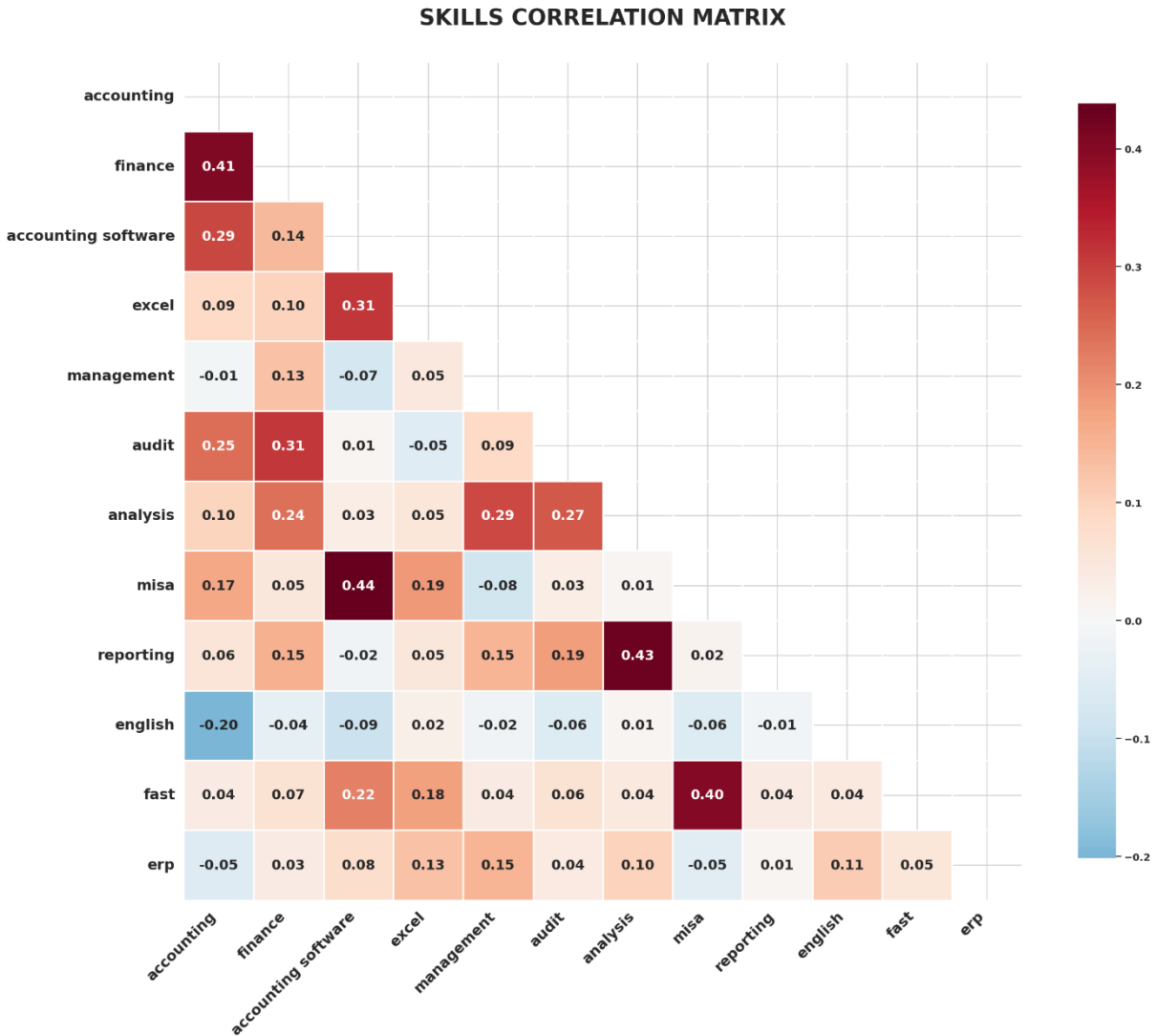


Figure 4.16. Skills Correlation Matrix

This correlation matrix provides a comprehensive view of the accounting and auditing job market, illuminating how technical tools, professional knowledge, and soft skills intertwine. One of the most prominent features of the matrix is the strong connection within the professional knowledge group. The correlation between Finance and Accounting is high at 0.41, while Finance and Audit also shows a notable link of

0.31, suggesting that audit roles are more closely associated with broad financial knowledge than with typical bookkeeping tasks.

Technically, the matrix emphasizes the critical role of specialized tools. MISA software shows a very high correlation with Accounting Software (0.44) and is also closely linked with Fast (0.40). Furthermore, the strong relationship between Reporting and Analysis (0.43) underscores that in the accounting field, the ability to generate reports is inseparable from the capacity to interpret data. The correlation between Excel and Accounting Software (0.31) further suggests that Excel acts as a vital bridge linking basic software input with the final output.

Regarding soft skills and language, there are some inverse or weak relationships. English proficiency shows a negative correlation with Accounting (-0.20), which may indicate a fragmented market where traditional accounting roles and those requiring high English proficiency are distinct segments. Additionally, while management is a key skill at higher levels, its correlation with technical tools like Accounting Software is slightly negative (-0.07), reflecting the shift from technical execution to strategic supervision as one advances to a Manager level (where salary peaks at 26.13 million VND).

In summary, the matrix confirms that the accounting job market is built on a "triple-threat" foundation: professional expertise (Accounting & Finance), technical proficiency (MISA, Fast & Excel), and analytical capabilities.

4.1.14. Pairplot of Key Numeric Variables

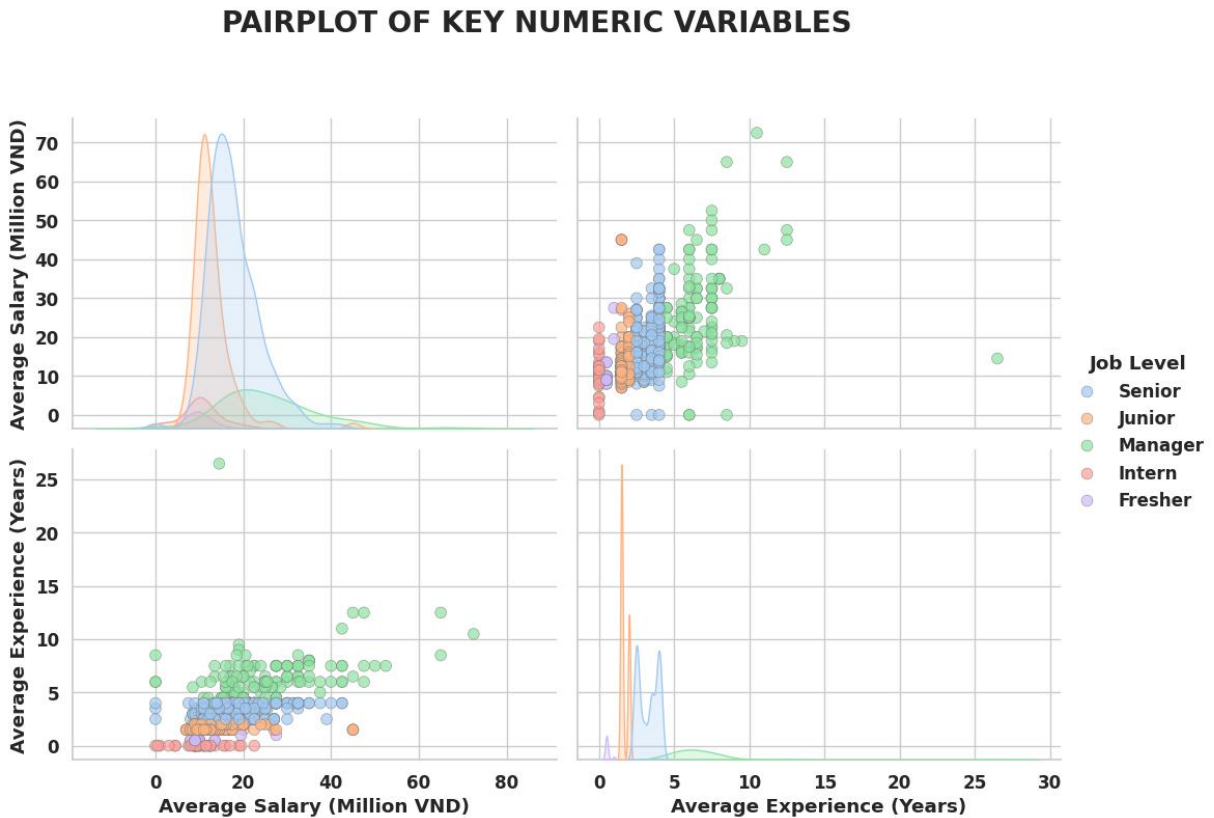


Figure 4.17. Pairplot of Key Numeric Variables

The density plot along the diagonal shows a large concentration of the labor market in the "low to mid" segment for both salary and seniority. The curves for Junior (orange) and Senior (blue) levels display the highest density peaks, with salaries typically ranging from 10 million to 25 million VND. This indicates a very high degree of market saturation and standardization for mid-level roles. Conversely, the Manager (green) curve shows a significantly elongated long-tail distribution to the right on the average salary axis. This suggests that while entry-level roles are constrained by the strict market average, management positions have high income elasticity, reflecting the diverse financial value assigned to strategic leadership capabilities, with average salaries reaching 26.13 million VND and some exceeding 60 million VND.

The graph further illustrates the vertical progression and hierarchical structure of the career path. At the starting point, the data points for Interns (red) and Freshers (purple) are very tightly clustered with near-zero variation in experience (0-1 year),

suggesting an extremely uniform starting point for newcomers. As the trajectory shifts to Junior and Senior, there is a clear overlap in the 2-5 year experience range. The Boxplot of Average Experience confirms that the median experience for the entire dataset is approximately 2.5 years, with the bulk of the market (interquartile range) falling between 1.5 and 4 years. This implies that in the Vietnamese context, the transition from Junior to Senior is not simply about accumulating years of experience but also requires specific performance milestones.

The Manager position occupies the upper right quadrant but displays a very wide horizontal dispersion on the salary axis. This variability highlights an income disparity, with some managers earning salaries comparable to senior-level roles (around 20-30 million VND), while others reach extreme levels exceeding 70 million VND. This suggests that once a Manager level is reached, income is no longer strictly dependent on the market average, but rather reflects individual negotiation skills and specialized abilities.

The chart shows a polarization between quantity and quality; while the workforce is abundant at the bottom (Interns and Freshers with average salaries of 10.26M and 11.31M VND), the highest incomes are reserved for a specialized minority at the top of the hierarchy.

4.2. Dashboard

The final experimental result for this study is a dashboard presenting an overview of the key findings analyzed from this dataset.

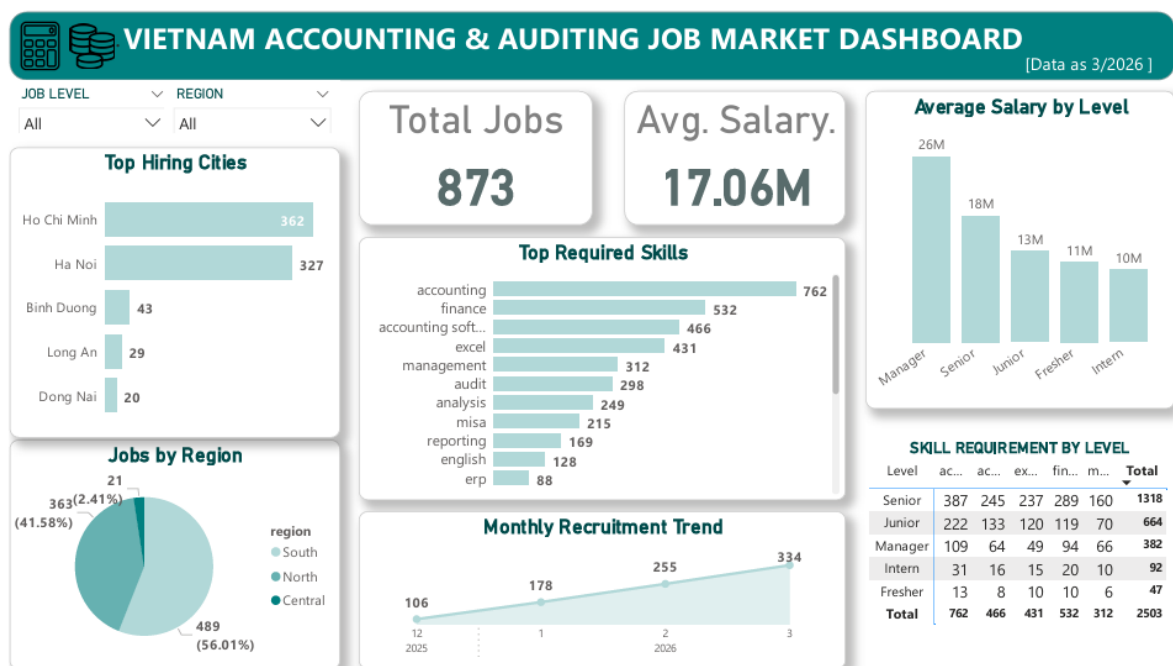


Figure 4.18. Dashboard of Accounting and Auditing Recruitment Market Analysis

This dashboard, developed with Microsoft Power BI, offers a detailed and interactive examination of the accounting and auditing employment landscape within Vietnam. The underlying dataset comprises 873 authentic job postings, gathered between December 2025 and March 2026, and encompasses the following elements:

Key Performance Indicators (KPIs): The analysis incorporates 873 pertinent job postings, which collectively present an average salary of VND 17.06 million.

Geographic Analysis: A pie chart showing the distribution of recruitment across all three regions (South, North, and Central), and a bar chart showing the top 5 localities with the highest recruitment demand.

Income Analysis: A bar chart illustrating the increase in average salary along a career progression path from Intern to Manager.

Skill Requirements: A horizontal bar chart shows the most in-demand hard and soft skills, and a matrix details the competencies required for different job levels.

Recruitment Trends: An area chart showing the fluctuations in recruitment demand from December 2025 to March 2026.

4.3. Discussion and Insights

Based on the visual analysis of 873 valid job postings up to March 2026, the accounting and auditing labor market in Vietnam reveals profound strategic trends and structural characteristics. Unpacking these data segments provides valuable insights for both employees and employers.

4.3.1. Geographical Polarization and the "Volume vs. Value" Paradox

The market exhibits considerable geographical disparity due to its concentration within major economic centers. The Southern region demonstrates a complete dominance in terms of volume, comprising 489 jobs and representing 56.01% of the market share, succeeded by the North with 363 jobs (41.58%), whereas the Central region constitutes a minimal fraction (2.41% with 21 jobs). This pronounced dominance of the South, especially Ho Chi Minh City (362 jobs) and its surrounding industrial provinces (Binh Duong, Long An, Dong Nai), indicates a substantial demand for data processing personnel to support foreign direct investment (FDI) enterprises and industrial zones.

However, the data reveals a compelling paradox: Although the South provides the most jobs, the North leads in average salary (17.667 million VND compared to 16.733 million VND in the South). This observation suggests that the Northern market, particularly Hanoi, which has 327 job postings, is prioritizing senior management roles, specialized advisory positions, and intricate auditing functions. These roles typically offer greater value and, as a result, more lucrative financial rewards.

4.3.2. Experience Barriers and the Dominance of Middle Management Teams.

The job landscape is dominated by experienced individuals. Senior professionals, with 440 positions, and Juniors, totaling 252, make up the bulk of the opportunities.

Opportunities for Interns and Freshers, however, are scarce. There are just 39 openings for Interns and a mere 18 for Freshers.

This sends a clear message: Enterprises are currently pursuing a strategy of "optimizing training costs." They are stringent with new graduates and prioritize candidates with "readiness"—the ability to immediately step in and handle tasks independently. Therefore, the greatest challenge for recent graduates is accumulating enough practical, hands-on skills to overcome the initial 1-3 year experience barrier.

4.3.3. Career Trajectory and the Salary Leap

The market's overall average salary stands at 17.06 million VND, adhering to a very strict seniority-based model in its early stages. During this initial phase, compensation is rather rigidly standardized: Intern (10.26 million VND), Fresher (11.31 million VND), and Junior (12.78 million VND). The small difference between these three levels suggests that the work's nature hasn't yet significantly changed in terms of its complexity.

However, a financial breakthrough becomes evident later in the career. Promotion to Senior elevates the salary to 17.79 million VND, representing a substantial increase of almost 40% over the Junior level. The Manager role, however, reaches an average salary of 26.13 million VND, which is 2.5 times the amount an Intern earns.

This extreme differentiation proves that the market is willing to pay massive premiums for risk management capabilities, a leadership mindset, and the ability to make strategic financial decisions.

The K-Means clustering analysis reveals a distinct “labor pyramid” structure, where the majority of the market (53%) consists of Intern and Fresher positions (Group 0) characterized by high turnover and entry-level pay. As professionals progress to Junior/Senior levels (Group 1) and eventually to Senior/Management roles (Group 2), the data shows a significant non-linear salary growth, with Group 2 earning nearly triple the entry-level average despite representing only a small, elite fraction of the workforce. This distribution underscores that while experience is a powerful driver for income appreciation, the path to high-paying management roles requires strategic skill upscaling

to transition out of the high-volume, low-wage segments and into the stable, high-value tiers of the market.

4.3.4. The Evolution of the Competency Framework: Expertise, Technology, and Ethics

Analysis from the Top Required Skills chart paints a portrait of a modern financial professional built on three core pillars:

Expertise is immutable: "Accounting" (762 counts) and "Finance" (532 counts) remain the paramount requirements, proving that a solid professional foundation is the mandatory ticket into the industry.

The standard for technology proficiency includes software skills, with accounting software (466 mentions) and Excel (431 mentions) being the most common. In Vietnam, domestic software like MISA, which is designed to meet local accounting standards, is used much more than global systems such as ERP (88 mentions) or SAP (215 mentions).

Employers increasingly value ethical behavior as a measurable skill, with traits like Carefulness (373 mentions), Honesty (381 mentions), and Responsibility (306 mentions) being highly sought after, almost as much as technical skills.

In a data-sensitive field, integrity and meticulousness are treated as non-negotiable professional competencies.

The English Language Gap: The data also indicates that possessing English language skills boosts a candidate's average income by approximately 14% (19.1 million VND vs 16.7 million VND), creating a significant competitive advantage for those in the "bilingual accountant" group.

4.3.5. Seasonality and the Post-Tet Recruitment Boom

The monthly recruitment trend chart illustrates a recovery and growth trajectory following a steep upward curve. From a trough of 106 job postings in December 2025 (a period when companies close their books and employees await year-end bonuses), the market rebounded strongly in the early months of the year, peaking at 334 postings in March 2026.

CHAPTER 5: CONCLUSION

5.1. Summary of the Study

This study primarily analyzes the Accounting and Auditing Recruitment Market using a dataset scraped from the CareerViet website. Data processing steps include handling missing values, dropping duplicates, and normalizing variables and text. The analysis and EDA are performed after the dataset is completely clean. Variables and their relationships are validated and visualized.

The output consists of dashboards containing essential information for the study, providing insights into the accounting and auditing job market, and offering important references, guidance, and key experience and skills relevant to job seekers and students.

5.2. Key Findings

Regarding recruitment needs:

- The labor market is heavily skewed toward Senior and Junior roles, which form the vast majority of recruitment demand.
- Opportunities for Freshers and Interns remain very limited, indicating a high entry barrier for newcomers in the accounting field.
- The South continues to be the primary hub for job volume, while the North remains highly competitive in terms of specialized and management-level compensation.
- Recruitment activity follows a clear seasonal cycle, showing a steady upward trend from the end of the year toward a significant peak during the opening months of the new year.

Regarding Salary Trends:

- Salaries show a consistent upward trajectory as candidates progress through job levels.
- Management positions command significantly higher compensation, reflecting the high value placed on leadership and strategic capabilities.
- Proficiency in English remains a key driver for income, providing a clear financial premium compared to roles with no language requirements.

Regarding skills:

- Accounting and Finance remain the non-negotiable foundations for all professional levels.

- Requirements shift from technical tool proficiency (Excel, Accounting Software) at staff levels to leadership and management thinking at higher positions.
- There is a strong emphasis on domestic software like MISA, showing a high correlation with practical reporting and analysis tasks.
- A "gap" exists for professionals who combine deep accounting expertise with strong foreign language skills, offering a significant advantage in the modern market.

5.3. Limitations

The biggest limitation of this study is that the data crawled from CareerViet is time-limited due to storage policies and the job posting refresh process, so only job information from the last four months is updated and displayed. Therefore, when analyzing job trends, factors related to time and number of jobs are not performed completely and logically because of the lack of information and reliance on the existing dataset.

Initially, keyword-based data collection generated a large sample size, which also resulted in a significant amount of duplicates. Coupled with the inability to handle missing values, such as replacing them with medians/means/modes (as this would significantly impact the mean), all missing and duplicate values were completely removed, leading to the loss of over 90% of the collected data.

The analysis only covered both accounting and auditing, not individual analysis or comparison between the two industries, because the text processing only standardized the job description and requirements, ignoring the title, thus failing to filter specifically for accounting or auditing.

5.4. Future Research Directions

Using this study as a foundation, and drawing lessons from the limitations, future directions will select a different website with a wider timeframe, identify key keywords to extract higher-quality data, or possibly scan multiple websites simultaneously to diversify the sample size. This will allow for a deeper analysis of the differences between sub-sectors within the same field, providing visual and objective insights so that job seekers and students can gather the most valuable information to create their own career paths.

Further expansion and upgrading will involve applying Machine Learning (ML) to research, using appropriate algorithms and models to predict trends, supply, and demand in the Vietnamese labor market.

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